Table of Contents

HARNESS AND CABLE ASSEMBLY SPECIFICATION

Document: WH-EEG-008 **Revision:** B **Date:** 2026-09-01 **Issued by:** TI One Voice research programme (one.witysk.org), Brussels, Belgium **Licence:** CC BY-SA 4.0 **Governing documents:** DSN-EEG-003 Rev D, then RFQ-EEG-001 Rev E. Where this document and design.py disagree, design.py governs.

Revision note, Rev A to Rev B. Re-issued against the 150.0 x 130.0 mm four-layer carrier and the enlarged POD-P1; WH-08, the captive host cable and its gland, is withdrawn and replaced by WH-09, the USB-B to USB-C panel pigtail; the boom preamplifier is moved to MP-01 and its part is stated as unsettled; the harness electrical limits are declared the single home for JIG-EEG-009 to cite; WH-BUS-01 is specified as a fabricable board in the new section 3.2.1 rather than as a line in a register; and the requirements this document cannot claim as met are now stated where the requirement is stated. **WH-KEY-01 and the five WH-ADP adapters are specified at this revision.** Rev A named all six in its wire lists, its build order and its test record and drew none of them, so five of the eight assemblies terminated on a part number with nothing behind it and the only mitigation against the one safety-relevant mis-mate in the kit had no geometry. The shroud and three of the adapters are modelled in `tools/mech_gen.py` and released as STEP and STL; the other two are bought parts and are specified by class. What each one still needs before it can be bought or printed is stated with it.

Corrections within Rev B, after the verification review of package v2.2 of 2 September 2026: the EMG lead colours are re-issued to the ruling. Section 3.6, its WH-06 lead table, the conductor-identification paragraph of section 10 and open item 7 of section 11 now carry **RD red / YE yellow / GN green**. Rev B as first issued bought those three leads as white, brown and grey and reported the difference with IFU-EEG-014 as undecided, which by then it was not: **IFU-EEG-014 Rev B section 13.2 rules the code red / yellow / green**, states the reasoning in its control note and records the ruling in its section 16 item 11, and PKG-EEG-015 Rev B section 4.2 already labels the sockets to it. A harness shop building to the earlier text would have fitted, in a code no other document uses, the three leads the participant is told to match by colour. The revision letter does not change; this is a correction within the same release, and it is not yet entered in the ECO-EEG-016 register, which is where the change record belongs.

Also within Rev B, 2026-09-02, after the completeness audit of package v2.3: the helmet-end and boom-end terminations are specified. New section 3.1.1 specifies the electrical joint at the eight HM-04 electrode assemblies, which Rev B named as an “HM-04 bayonet tag” and which exists as no part, no feature and no method; new section 3.1.2 resolves the two ear-reference terminations, which this document, AVL-EEG-017 K2 and SVC-EEG-013 R4 described as three mutually incompatible joints. Section 4 gains a WH-03B column, so the 1700 mm boom lead has a construction for the first time. Section 6 states **one** crimped 2.54 mm connector system for the whole kit and says why it is that one. Section 8 gains steps 14a and 14b. AVL-EEG-017 section 4 now carries the purchasing lines that the materials, connectors and tooling of sections 4 and 6 never had. **Sections 3.1.1 and 3.1.2 are PROPOSALS and are marked as such in their own headings**: they need a mechanical reviewer, a safety reviewer and a vendor, and none of the three has answered. The revision letter does not change; this is a correction within the same release, and it is not yet entered in the ECO-EEG-016 register.

Also within Rev B, 2026-09-02, after the decision review of package v2.3: the three helmet terminations and the electrode preload spring are ruled. Rulings D2-EAR-REFERENCE-COUPLER, D3-BIAS-FPZ-TERMINATION and D5-K12-SPRING-ENVELOPE are written into sections 1, 3.1, 3.1.1 to 3.1.4, 4, 6, 7, 8, 9, 10, 11 and 12 at this issue. Section 3.1.2 keeps the free-hanging touch-proof coupler and gains the provisions the ruling attaches to it, of which the load-bearing one is a **packing rule this document does not own**: the ear clips travel mated and captive, or the participant mates and unmates two of them every session out of a bay of five identical 1.5 mm DIN plugs, and the safety argument that section rests on is not true. New section 3.1.3 terminates conductor 11 at a free-hanging touch-proof socket on the halo front and **deletes the “Fpz bias pad” as a helmet feature**, which was open item 26. New section 3.1.4 **issues** the K12 spring specification, which ruling D5 declined to issue against the 4.50 mm spring seat it was written on and which is derivable now that `tools/mech_gen.py hm04()` has deepened that seat to 6.60 mm. **None of the three is a signature.** Each names who must sign it and section 11 carries those names.

And the released HM-04 and HM-05B changed under this document on the same day, which is why sections 3.1.1, 6, 7 and 9 are corrected against them rather than left standing: the bayonet now turns (entry slots 1.70 mm wide at outer radius 5.55 mm cut 3.60 mm deep, a 100 degree circumferential run at z 1.10 to 3.80 and a 1.10 mm retaining lip below it, with the HM-05B lug corrected from r 5.40 to the r 5.20 its own docstring always claimed); the single through-window is now **two** pockets with 1.60 mm of PA12 between them, so an LED lead and an electrode conductor no longer share one cavity, which was RISK-EEG-011 SF-9; and the spring seat roof moved from z 13.50 to z 15.60. The revision letter does not change; this is a correction within the same release, and it is not yet entered in the ECO-EEG-016 register.

Why this document exists

In package v1 the helmet harness existed only as a prose paragraph in DSN-EEG-002 Rev E section 6 and as unpublished net names inside a KiCad board file. Four documents gave four different conductor counts for the same cable – 20-way, 21-way plus shield, 22 positions, and a 27-conductor bundle in a 20-way FFC – and none of them added up. No harness shop can cut wire from that, no contract manufacturer can quote it, and no goods-inwards inspector can check what arrives. ECO-EEG-014 has since split the harness into two cables so that eight switched digital lines no longer cross the analogue zone of the carrier, which changes the arithmetic again. This document is the single from-to authority for every cable in the kit: eight assemblies, every conductor, every pin, the materials, the connectors and the tooling, the routes inside HM-01, the build order, the tests and the record. Nothing in it has been built or measured, and no safety engineer has reviewed this design. Every dimension, resistance and fill figure below is **calculated** from `tools/design.py`, from the HM-01 and POD-P1 geometry, and from supplier catalogue data, and is labelled as such.

---

1. Cable register

Nine controlled assemblies at this issue *(was eight; WH-10 is added on 2026-09-02 by section 3.1.3, and like WH-06 it is bought in rather than built)*. Part numbers are WH-EEG-008-nn; the short form WH-nn is used throughout this document and on the labels. Carrier connector coordinates are given in the `design.py` convention (top-left origin, Y down) and in the fabrication convention (bottom-left origin, Y up, `y_out = 130.0 - y_design`), because the two appear on different drawings and confusing them puts a connector on the wrong edge of the board.

Ref	Assembly	Ways	Carrier connector	Position, design (mm)	Position, fab (mm)	Screened	Qty/kit
WH-01	Helmet electrode cable	12	J14, 1x12 socket strip	5.0, 12.0	5.0, 118.0	yes, overall foil + drain	1
WH-02	Helmet contact-light cable	10	J30, 1x10 socket strip	66.0, 90.0	66.0, 40.0	no – see section 5.4	1
WH-03	Boom microphone pigtail	4	J18, 1x4 socket strip	122.0, 90.0	122.0, 40.0	yes, overall foil + drain	1
WH-04	Headphone panel pigtail	4	J27, 1x4 socket strip	128.0, 72.0	128.0, 58.0	yes, overall foil + drain	1
WH-05	Room microphone cable	4	J28, 1x4 socket strip	122.0, 102.0	122.0, 28.0	yes, overall foil + drain	1
WH-06	EMG DIN lead set	1 each	J15, J16, J17 (DIN 42802)	8.0, 76/88/100	8.0, 54/42/30	no – see section 3.6	3
WH-07	Charge-port pigtail	2	J24, JST PH 1x02	143.0, 80.0	143.0, 50.0	no	1
WH-09	Isolator host pigtail, USB-B to USB-C panel receptacle	4 + shell	none – module to panel	(module at J10: 136.0, 6.0)	(136.0, 124.0)	cable's own braid	1
WH-10	Fpz bias lead, snap to DIN 42802 touch-proof plug, 150 mm	1	none – mates the WH-01 conductor 11 coupler on the helmet	–	–	no – see section 3.1.3	1

WH-08 is withdrawn and its number is not reused. Rev A specified WH-08 as a captive 2 m host USB-C cable leaving the pod through a Lapp SKINTOP gland. The host connection is a socket, not a captive cable: the host connector is the USB receptacle on the ADuM4160 isolator module, presented through a gasketed aperture in POD-P1, and the participant plugs one of the two A-07 cables into it. There is no gland in the Phase 1 build and no gland feature exists in POD-P1-01. A captive lead through a gland remains a Phase 2 candidate for the helmet shell and is not specified here. Section 3.8 specifies what replaced it.

Two sub-assemblies are called out because they carry conductors and would otherwise fall between documents:

Ref	Assembly	What it is
WH-BUS-01	Contact-light bus board	14.0 x 10.0 x 0.80 mm FR-4, ten 1.60 mm plated pads on a 2.70 x 4.80 mm grid, no components. Sits at frame node N1 immediately inside the occipital entry. Splits LED_V into eight tails without a single crimp splice. Fabrication data is `kicad/wh-bus-01/`, generated by `tools/wh_bus.py`; section 3.2.1 specifies it. **Two copper layers, not one**, for the reason in 3.2.1; PARTS-EEG-019 is corrected to it at its 2026-09-02 issue.
WH-03B	Boom lead	Capsule to a 4-pole 3.5 mm plug, 1700 mm. Two 7/0.1 mm PTFE conductors, overall foil and 30 AWG drain, TPU jacket OD 2.2 mm maximum. Part of the boom assembly. **Its construction is section 4, which gains a WH-03B column at this issue**; its wire list is section 3.3; AVL-EEG-017 K41 buys it. The “TRRS plug” of the first issue is the same 4-pole part and the word is dropped, because the fourth contact here is a screen return and not a microphone ring.
WH-KEY-01	Keying shroud	Printed PA12 tube that drops over a carrier socket strip and takes the harness housing in one orientation only. Three forms, for J14, J30 and J22. Dimensioned in section 6.1; geometry in `tools/mech_gen.py wh_key01()`.
WH-ADP-01, -01B, -02, -03, -04	Panel adapters	The five parts the panel-going pigtails terminate on. Two are bought jacks, three are printed. Specified in section 3.9.

No cable in this register lands on J22. J22 is the 1x3 socket carrying the two spare protected electrode channels EOGIN1 and EOGIN2 and their AGND_REF screen. The EOG panel sockets, their cable and their drawing are a Phase 2 option, listed as such in PARTS-EEG-019 with no part number yet, so in a standard build channels 15 and 16 are protected and reach no panel socket. That is a stated omission, not an oversight.

Phase note. Phase 1 puts the electronics in the POD-P1 bench enclosure (163.0 x 143.0 x 58.0 mm external, 158.0 x 138.0 x 55.5 mm internal, with the 163.0 x 143.0 x 6.0 mm lid), so WH-01 and WH-02 are 1500 mm umbilicals from the helmet to the pod, and the boom plugs directly into the POD-P1 panel rather than running through the frame. In

Phase 2 the electronics move into the occipital shell of HM-01 and both umbilicals shorten to 180 mm; WH-03 then routes through the left temple channel as DSN-EEG-002 section 6 describes. The Phase 2 shell is sized in RFQ M-01 for a carrier that no longer exists at that size, so the -H2 variants cannot be dimensioned yet. The two variants are WH-01-P1 / WH-01-H2 and are not interchangeable. Only the -P1 variants are being built.

1.1 What changed on the carrier, and why this document had to be re-issued

Two findings came out of actually laying the board out rather than asserting that it would lay out, and both reach this document through the connector coordinates and the keep-out.

The carrier grew from 130 x 124 mm to 150.0 x 130.0 mm. Thirty connectors, 211 parts and 156 nets would not close at the smaller size. At kit quantities the extra bare board costs a few euro per unit against a real risk of an unroutable design. Every carrier coordinate in this document, and the `y_out = 130.0 - y_design` conversion, follows from that: a harness built to the Rev A coordinates would be cut to the wrong service lengths and the fabrication drawing would put J14 on the wrong edge.

The carrier went from two layers to four. Package v1 argued that a two-layer carrier is cheap and easy to route. Doing the layout showed that it is not: on two layers the bottom side has to be both the reference plane and the second routing surface, and it cannot be both. The stack is now L1 signal, L2 reference plane, L3 reference plane, L4 signal, mask / 35 um L1 / prepreg 0.200 / 17 um L2 / core 1.065 / 17 um L3 / prepreg 0.200 / 35 um L4 / mask, 1.60 mm +/- 10 %, with through vias only, 0.60 mm pad on a 0.30 mm finished hole. The reference planes are AGND_REF left of x = 62 mm and DGND right of it, on **both** inner layers, stitched together only at the star points. That matters to the harness in three places: every electrode conductor now lands on a socket with a continuous reference plane under it rather than a swiss-cheesed pour, which is the condition section 5 assumes when it allows the in-frame runs to be unscreened; the isolation keep-out at section 3.8 is a keep-out on four layers, not two; and the two star points of section 5.1 are the only ties between two planes that now exist twice over.

---

2. Conductor arithmetic

The v1 statements and what Rev B actually implements:

Source	Claim	Status
DSN-EEG-002 Rev E s6, table W1	8 x screened pair + 2 x screened + 1 x screened + 8 LED = 27 conductors	Superseded. “Pair” meant signal plus its own screen, never signal plus return.
RFQ-EEG-001 Rev C, E-09	“J14, 20-way plus shield”	Superseded by ECO-EEG-014; E-09 is re-issued in RFQ-EEG-001 Rev E as two cables.
EEG-CAR-01_BOM.csv, J14	“Helmet harness (20-way + shield)” on a 1x22 footprint	Superseded. J14 is 1x12 in Rev B.

Source	Claim	Status
Kit BOM Rev B, row 24	“20-way FFC to carrier”	Wrong construction. An FFC has no per-conductor screen, no drain and cannot mate a 2.54 mm socket strip. Delete “FFC”.
ECO-EEG-016, ECO-EEG-014 “Was” text	“eleven electrode conductors and **eleven** contact-light conductors” in one 22-way socket	Wrong on the light side. The light group is ten conductors, not eleven: eight LEDn plus LED_V plus LED_GND. The budget below is the one that adds up and ECO-EEG-016 is corrected to it.

Rev B budget, which does add up:

Group	Conductors	Cable	Connector
Eight scalp electrodes	8	WH-01	J14.1 to J14.8
Two ear references	2	WH-01	J14.9, J14.10
Bias drive to Fpz	1	WH-01	J14.11
Commoned screen drain	1	WH-01	J14.12
Eight contact-light lines	8	WH-02	J30.1 to J30.8
Light common and guard	2	WH-02	J30.9, J30.10
Total into the helmet	**22**	two cables	two connectors

The count is unchanged at 22. What changed is that the eleven high-impedance electrode conductors and the eight switched 3.3 V lines are now in separate jackets, entering the carrier 61 mm apart on opposite sides of the zone split at x = 62 mm, instead of sharing one socket at x = 5 mm in the middle of the analogue zone.

The kit terminates on **fourteen** places on a person: eight scalp cups, two ear references, one bias pad and three EMG studs. **The bias pad is a disposable pre-gelled snap electrode off the K4 pack from 2026-09-02**, placed on prepped forehead skin below the HM-02A brow pad and replaced every session, not a feature of the helmet – section 3.1.3. The count is unchanged; what changed is that the fourteenth termination now exists as a part. The two EOG spares are protected on the carrier and are not fitted in a standard build.

---

3. From-to wire lists

Colour names are IEC 60757 two-letter codes. **Solid** colours belong to WH-01 (electrode); the same colour **with a black tracer** belongs to WH-02 (light) for the same site, so a technician with the crown cover strip off can tell a Cz electrode conductor from a Cz light conductor by eye. Black is used only for 0 V conductors, and never in WH-01, because the bias lead is a driven common-mode return and not an earth.

3.1 WH-01 – helmet electrode cable, 12-way screened, to J14

Insulation PTFE throughout. Gauge 7/0.1 mm tinned copper for all eleven signal conductors. Cut lengths are calculated for the -P1 variant: 1500 mm umbilical + 120 mm pod service loop + 20 mm pod termination allowance + the in-frame routed length + 40 mm site service loop + 15 mm site termination allowance. Tolerance +10 / -0 mm.

Conductor 11 is the one conductor that does not follow that formula, from 2026-09-02. Its site is a free-hanging coupler at the halo front and not a body the harness is dressed into, so there is nowhere to coil a 40 mm site service loop; the loop is deleted and a stated free tail **F** takes its place, giving 1500 + 120 + 20 + 285 + 15 + F = **1940 + F mm**. F is a fitting dimension and is set at the first fitting trial – open item 30. *The 1980 mm this row carried before that date is superseded: the arithmetic was right for a conductor soldered to a fixed pad, and it contains no free tail at all, so it cannot also be a free-hanging coupler.*

Cond	Colour	Gauge	Insul.	From (site / terminal)	To (pin)	Net	Carrier path	In-frame (mm)	Cut (mm)
1	WH white	28 AWG eq.	PTFE	Fz cup, HM-04 termination (3.1.1)	J14.1	E_Fz	R1 47k -> D1 -> C1 -> IN1 (J2.1)	315	2010
2	RD red	28 AWG eq.	PTFE	Cz cup, HM-04 termination (3.1.1)	J14.2	E_Cz	R2 -> D2 -> C2 -> IN2 (J2.2)	205	1900
3	BU blue	28 AWG eq.	PTFE	Pz cup, HM-04 termination (3.1.1)	J14.3	E_Pz	R3 -> D3 -> C3 -> IN3 (J2.3)	130	1825
4	GN green	28 AWG eq.	PTFE	C3 cup, HM-04 termination (3.1.1)	J14.4	E_C3	R4 -> D4 -> C4 -> IN4 (J2.4)	285	1980
5	YE yellow	28 AWG eq.	PTFE	C4 cup, HM-04 termination (3.1.1)	J14.5	E_C4	R5 -> D5 -> C5 -> IN5 (J2.5)	285	1980
6	BN brown	28 AWG eq.	PTFE	T7 cup, HM-04 termination (3.1.1)	J14.6	E_T7	R6 -> D6 -> C6 -> IN6 (J2.6)	175	1870
7	OG orange	28 AWG eq.	PTFE	T8 cup, HM-04 termination (3.1.1)	J14.7	E_T8	R7 -> D7 -> C7 -> IN7 (J2.7)	175	1870
8	VT violet	28 AWG eq.	PTFE	F7 cup, HM-04 termination (3.1.1)	J14.8	E_F7	R8 -> D8 -> C8 -> IN8 (J2.8)	250	1945
9	GY grey	28 AWG eq.	PTFE	Left ear clip, A1 coupler: free-hanging 1.5 mm DIN 42802-1 socket, GY body (3.1.2)	J14.9	REF_L	R9 -> D9 -> C9 -> SRB1 (J2.9)	195 + 250 free	2140
10	PK pink	28 AWG eq.	PTFE	Right ear clip, A2 coupler: free-hanging 1.5 mm DIN 42802-1 socket, PK body (3.1.2)	J14.10	REF_R	R10 -> D10 -> C10 -> SRB1 (J2.9)	195 + 250 free	2140
11	TQ turquoise	28 AWG eq.	PTFE	Fpz, A3 coupler: free-hanging 1.5 mm DIN 42802-1 socket at the halo front, TQ body (3.1.3)	J14.11	BIAS_EL	R11 from BIASOUT (J2.10); **D11, C11 on BIASOUT**	285	**1940 + F** (3.1.3)

Cond	Colour	Gauge	Insul.	From (site / terminal)	To (pin)	Net	Carrier path	In-frame (mm)	Cut (mm)
12	bare, clear sleeve	30 AWG drain	–	commoned screen, pod end only	J14.12	HARN_SHIELD	**R91 (0R) to DGND – the only connection**	none	1660

D1 to D16 are **BAV99**, not BAT54S. Schottky leakage across a 47 kOhm series resistor is an offset error on a 10 uV input; BAT54S is used only at the envelope rectifiers D20, D40 and D60. This is stated here because the harness operator sees the clamp designators on the board and may otherwise assume they are all the same part.

Two facts a builder must be told, because neither is visible in the cable:

- **REF_L and REF_R are hard-paralleled on the carrier.** R9 and R10 both land on net SRB1 (J2.9, J4.9), so the two ear clips sit in parallel through 2 x 47 kOhm = 23.5 kOhm to the shared reference. There is no separate ear channel and one clip falling off is electrically almost invisible. The harness carries two conductors, not one, and both must be present and continuous.
- **The bias path runs outward.** BIASOUT leaves the module at J2.10, passes through R11, and only then becomes BIAS_EL on the cable. Conductor 11 is therefore a driven output, not an input, and the resistor is upstream of the person exactly as it is on every other lead (RFQ E-07). **Corrected 2026-09-02:** the clamp and the capacitor are upstream of the person now as well. `tools/design.py` lost the channel-11 override at this issue, so the row is generated by the same loop as the other fifteen and D11.3 and C11.1 land on BIASOUT, behind the 47 kOhm. *Was: "D11, C11 on BIAS_EL", which put both on the patient side of R11 and made this the one lead of sixteen whose single-fault current its own series resistor did not bound.* What that changes, and what it does not, is stated in section 3.1.3 beside the current numbers; where this document and `design.py` disagree, `design.py` governs.

3.1.1 The termination at HM-04 – a PROPOSAL, not a released design

Nothing in this section is released. It specifies a joint that does not exist, asks for geometry that two released models do not have, and names two parts that have no vendor. It is written here because the alternative is what Rev B shipped: eight conductors landing on an "HM-04 bayonet tag" that is on no drawing and in no model, and a build step (section 8, step 14) that defers to an ASM-EEG-007 step which does not exist.

What is actually missing. Conductors 1 to 8 of WH-01 and both leads of all eight contact lights end inside an HM-04 body: eight electrode conductors and sixteen LED leads, **twenty-four joints per helmet, none of which has a terminal, a method or a wire entry.** ASM-EEG-007 Rev B stage 3 bonds the bodies (4.2), fits the LEDs (4.3), threads the harness (4.4) and fits the yoke, strap, boom and pads (4.5); it has no termination step, no step that fits the cups, the bayonet carriers or the springs, and no stage-3 sign-off line for either. MECH-EEG-020 sheet 8 shows a cup bore, two bayonet slots, a gel port and a side window, and no terminal, no tag, no wire entry and no LED seat. So the site end of both helmet cables cannot be made, and test H6's "HM-04 termination, 15 N minimum" has no joint to pull.

Three released files already say three different things, which is the reason this is a decision and not a drafting job:

Source	What it says
This document, section 3.1 as first issued	“Fz cup, **HM-04 bayonet tag**” – a feature of HM-04
`tools/mech_gen.py hm05b()`	“a 1.60 x 0.80 mm slot in the flank for the **WH-01 conductor’s solder tag**, which WH-EEG-008 H6 pull-tests at 15 N” – a feature of **HM-05B**, the part that turns and comes out
ASM-EEG-007 Rev B stage 3	nothing at all

The second is real geometry: the released `mech/stl/HM-05B_cup_bayonet_carrier.stl` carries a blind pocket 1.60 mm wide, 1.20 mm deep into the flank and 0.80 mm tall at z = 5.60 to 6.40, at 90 degrees to the bayonet lugs *(read from the mesh)*. It is the only feature anywhere in the package that was drawn for this joint, and **it cannot be the answer**, for two reasons that no drawing note can fix.

- **Nothing can reach it.** HM-05B’s body is 9.10 mm in HM-04’s 9.20 mm bore. The pocket sits at z = 5.60, mid-bore, enclosed by 0.05 mm of radial clearance for its whole circumference. A conductor soldered into it has no route out of the assembly.
- **It is on the part that leaves.** SVC-EEG-013 section 3 R4 releases all eight cups with the HM-09 key at every turnaround, R5 puts them in a 40 kHz ultrasonic bath, and R10 refits them. A conductor soldered to the carrier is either unsoldered eight times per turnaround or taken into the bath with the frame, the harness and 1500 mm of umbilical attached to it.

What the released HM-04 has. Read from `mech/stl/HM-04_electrode_assembly_body.stl` and `tools/mech_gen.py hm04()`, cup face at z = 0, gel-port face at z = 18.00. *(model figures; no HM-04 has been printed)*

Feature	Geometry
Body	12.40 x 12.40 x 18.00 mm square prism, 1.90 cm3, top edges filleted 1.20 mm
Cup bore	9.20 dia, z 0 to 9.00, open at the cup face
Spring seat	6.80 dia, z 9.00 to **15.60** *(was z 9.00 to 13.50; deepened 2026-09-02, section 3.1.4)*
Gel port	2.50 dia on the axis, through the top face into the cup bore
Bayonet entry slots	two, 2.30 radial x **1.70** tangential, **3.60 mm** deep from the cup face, outer radius **5.55 mm**, 180 degrees apart *(was 2.20 x 1.40 mm, 2.40 mm deep at 4.20 mm radius – a 1.40 mm slot for a 1.40 mm lug, in a material printed to +/-0.15 mm)*

Feature	Geometry
Circumferential bayonet run	**100 degrees**, r 4.30 to 5.55, z 1.10 to 3.80, both slots, with a **1.10 mm retaining lip** below it. *New on 2026-09-02. Before that date there was no run and the carrier was a plug fit*
Two side pockets	each 3.20 mm wide x 2.60 mm tall, z 11.70 to 14.30, 5.40 mm deep from opposite faces, **separated by 1.60 mm of PA12 on the axis**: outboard is the LED seat, inboard is the electrode conductor's run. *(Was one 3.20 mm box cut right through the body. The correction is dated 2026-09-02 and its reason is RISK-EEG-011 SF-9 – see below)*
Anything else	none. The rest is solid PA12

Two properties of that geometry decide what follows.

The window is now two pockets, and that is a safety correction and not a drafting one. *Superseded 2026-09-02. Was: "The side slot is a through-slot, not a blind window. `hm04()` cuts a 3.20 x 14.40 x 2.60 mm box through a 12.40 mm body, so the part has two openings of 3.20 x 2.60 mm at 180 degrees… The outboard opening is the light window the LED of ASM section 4.3 sits behind. The inboard opening is unused, and it is the only way into the assembly the part has."* That reading was correct against the model of the day, and the model of the day was the fault: one cavity through the body put a switched LED lead and a high-impedance electrode conductor about 3 mm apart, in shared open volume, at all eight sites, which is RISK-EEG-011 SF-9 written as geometry. `hm04()` now cuts **two** pockets of 3.20 x 2.60 mm, 5.40 mm deep from opposite faces, with 1.60 mm of printed PA12 between them: **the outboard pocket is the LED seat and the inboard pocket is the conductor run**, and they are no longer two ends of one hole. **ASM-EEG-007 section 4.2's fitting note, "the window is on one side of the body only", still does not match the released model** – there are two openings, they are simply no longer connected – and that correction remains ASM-EEG-007's.

There is no other route in. HM-05B's body is 9.10 mm in the 9.20 mm bore and its spigot 6.60 mm in the 6.80 mm seat – 0.05 and 0.10 mm on the radius – so nothing passes either annulus, and the 2.50 mm gel port is committed to gel. The free volume inside a seated assembly is the **3.50 mm** the seat now has above the spigot: HM-05B is 12.10 mm tall (`hm05b()`, and `mech/MANIFEST.json` bounding box 10.80 x 9.10 x 12.10 mm) and the seat roof is at **15.60**. *Was 1.40 mm against a seat roof at 13.50, until `hm04()` deepened the seat on 2026-09-02.* The inboard pocket, at z 11.70 to 14.30, opens into the lower 2.60 mm of that band, and section 3.1.4 spends 2.10 mm of it on the spring at solid height.

What the joint has to do. Five things, and between them they rule out every simple answer.

1. **Part at every turnaround, without a tool.** SVC-EEG-013 section 3 R4, R5 and R10, eight times each way, for the life of the kit.
2. **Not become a second half-cell.** The measurement is microvolts from a sintered Ag/AgCl surface. A junction of two different metals in the wetted path is an electrochemical cell with its own drift; a junction of two different metals anywhere

is a thermocouple in a 10 uV measurement. So: the same plating on both mating faces, and out of the gel.
3. **Survive 15 N** (test H6).
4. **Be re-makeable in the field.** SVC-EEG-013 section 5.3 replaces one conductor by drawing a new one through the channel and terminating it at the HM-04, with a soldering iron and hand tools, without disturbing the other seven sites.
5. **Not be a connector handled eight times per turnaround with wet gloves.** Whatever parts at the bath, the operator's hands are already busy with the bayonet and a released cup.

The proposal: the bayonet is the disconnect.

Make the electrical interface an **annular contact at the top of the spring seat**, so that the quarter-turn that already releases the cup is also what parts the circuit, and no operator ever plugs or unplugs anything at a site.

Element	Part	What it is
Rotating half	**HM-05C**, cup contact crown	An annulus let into the top face of HM-05B's spigot, flush at z = 12.10, clear of the 2.40 mm gel passage, gold over nickel on its upper face. Its tail runs to the cup down a dry groove in the carrier wall
Fixed half	**HM-04A**, electrode termination contact	A sprung leaf anchored in HM-04, reaching into the 1.40 mm band above the spigot and bearing axially on the crown, gold over nickel on its bearing face. Its tail leaves through the **inboard** side-slot opening and is soldered to the WH-01 conductor 8 mm outside the body
Spring	K12, **specified at section 3.1.4 from 2026-09-02**	Stays purely mechanical and carries no signal. *The words "it bears on the crown" are superseded on 2026-09-02: the spring bears on the PA12 spigot rim outboard of the crown and on the PA12 seat roof, and touching the gold crown is the thing section 3.1.4's no-gold relation exists to prevent. At the crown outside diameter this document proposes today, that relation does not close – see 3.1.4*

Being an annulus, the crown is **rotation-invariant**: the carrier turns 90 degrees to lock and the contact does not move on it, so nothing twists and no conductor follows the bayonet. Being axial, the contact is closed by the same 3 to 6 N the spring already applies to hold the cup against the scalp, so **a cup that is seated is by construction a cup that is connected**, and the failure this design must not have – a cup that looks fitted and is open – becomes a cup that is visibly not locked.

Two figures a contact supplier has to meet, and only one of them can be derived here.

- **Contact force at the crown: 0.5 N minimum.** *This is stated as a requirement. It is not derived from a measurement and no contact has been tested.* A gold-to-gold interface on a high-impedance input behind a 47 kOhm resistor has no fritting current available to break a surface film, so the normal force is the only thing that makes the contact.
- **Contact resistance is not the specification, and saying so matters.** The source impedance at this node is 5 to 20 kOhm of skin contact plus the 47 kOhm series resistor. A whole ohm of contact resistance is 14 parts per million of it, exactly as section 4 says of the conductor. What matters is that the contact is closed and stays closed. H1’s 1.0 Ohm limit is there to catch an open joint, not to protect the measurement.

What the released models have to gain. None of it is this document’s to draw.

Part	Feature the proposal needs
HM-04	an anchorage for HM-04A that takes 15 N, reachable from the inboard face. **Re-stated 2026-09-02:** the material it was to be cut in is no longer a full section. The spring seat now runs to z 15.60, so between the pocket roof at z = 14.30 and the top fillet at z = 16.80 the part is an annulus outside r 3.40 and not 2.50 mm of solid PA12. The anchorage and the deepened seat have to be drawn against each other
HM-04	a dressed exit at the inboard pocket – a radius, not a printed edge, against a 0.70 mm OD PTFE conductor
HM-04	~~a seat and two lead passages for the contact-light LED at the outboard opening~~ **– the seat is cut. Closed 2026-09-02:** `hm04()` cuts the outboard pocket as the LED seat, separated from the inboard conductor run by 1.60 mm of PA12. The **two lead passages out of that pocket are still owed**, and until they exist ASM section 4.3 threads two LED leads through a blind pocket
HM-04	~~the circumferential bayonet run~~ **– cut. Closed 2026-09-02:** 100 degrees at r 4.30 to 5.55, z 1.10 to 3.80, over a 1.10 mm retaining lip, against an HM-05B lug corrected to outer radius 5.20 at z 1.20 to 3.30. `tools/simulate_production.py` measures 0.000 mm3 of interference through the quarter turn and through the carrier’s 0.40 mm of axial travel, and 1.557 mm3 of lip engagement on a straight pull. That is a boolean measurement on the released solids, not a print: no HM-04 has been printed
HM-05B	a 0.50 mm recess in the spigot top for HM-05C and a dry groove down the carrier wall for its tail. **The 1.60 x 1.20 x 0.80 mm flank pocket is deleted or re-cut as that groove’s anchorage**, and `hm05b()`’s docstring reference to a WH-01 solder tag is withdrawn with it

Part	Feature the proposal needs
HM-05A	a defined tail. The cup as bought carries a 1.5 m lead (AVL-EEG-017 K1) and this joint wants tens of millimetres – see the note on K1 in that document
MECH-EEG-020	a sheet 8 that dimensions all of the above

What was considered and rejected.

Considered	Rejected because
The spring is the conductor	It is 302 stainless (AVL K12) sitting in a seat on the gel path. Stainless against Ag/AgCl in saline is a drifting half-cell in series with the measurement, and making the preload member the contact means every hair under a cup is an electrical fault as well as a mechanical one
A wiping contact in the cup bore	The bore is the gel volume: the port delivers into it and SVC R6 flushes it. A contact there is wetted by design
The `hm05b()` flank solder tag, as drawn	No route out of the assembly, and it is on the part that goes in the ultrasonic bath. Both are stated above
A miniature plug and socket at the site	3.20 x 2.60 mm of opening, eight sites, mated and unmated with wet gloves at every turnaround, with a released cup hanging off the connector while it is done
Solder the conductor to the cup and put the frame in the bath	R5’s bath is 40 kHz with an enzymatic cleaner and SVC-EEG-013 does not immerse the frame. It also turns SVC section 5.4’s ten-minute cup change into a harness repair
Buy the cup as a complete leaded electrode and put the joint in the frame channel	**This is the fallback and it is not a straw man.** It needs a passage out of HM-04 that the part does not have, a connector small enough to live under a 3.80 mm channel cover, and a way to get a 1.5 m factory lead down to a tail. The mechanical reviewer should weigh it against the crown

Who has to decide. The **mechanical reviewer**, because every row of the “must gain” table is a change to a released model and to MECH-EEG-020 sheet 8. The **safety reviewer**, because this is a joint in the patient-applied path and RFQ S-02’s single-fault case already does not close (section 5.5). **AVL-EEG-017**, because neither HM-04A nor HM-05C has a vendor: both are carried there as OPEN WITH CRITERIA, on lines K25 and K26, with the criteria above. Until all three have answered, the site end of both helmet cables cannot be built. That is open item 22, and the missing ASM-EEG-007 steps are open item 28.

Conductor 11 is not solved here. BIAS_EL lands on an “Fpz bias pad, solder tag” which is not an HM-04, is on no drawing and has no part number. It has exactly the defect this section opened with and this section does not close it – open item 26.

3.1.2 The two ear-reference terminations – also a PROPOSAL

Three documents describe three incompatible joints for the same two terminations, and a builder cannot choose between them.

Source	What it says	What it implies
Section 3.1 as first issued	“Left ear clip, **A1 crimp**”, on 250 mm of free conductor out of the temple	the harness conductor is crimped onto a **bare clip**: a permanent joint, onto a part that is not a catalogue item
AVL-EEG-017 K2	“Ag/AgCl ear-clip reference electrode, **DIN 42802 plug**”, two per kit	a **finished leaded electrode** with a touch-proof plug – and there is nowhere to plug it in. J15, J16 and J17 are the three EMG channels and carry EMGIN1 to EMGIN3
SVC-EEG-013 section 3 R4 and R5	“Release the two ear clips”, then the ultrasonic bath	the joint must part **without a tool**, twice per turnaround, for the life of the kit

A permanent crimp cannot satisfy the third. A plug with no socket cannot satisfy the second.

The proposal: a free-hanging touch-proof socket at each temple. Conductors 9 and 10 leave the left and right temple as the 250 mm free tails section 3.1 already gives them, and each is terminated in a **free-hanging 1.5 mm touch-proof socket to DIN 42802**. The ear clip is bought exactly as AVL-EEG-017 K2 already buys it – complete, leaded, with its own touch-proof plug – and plugs into that socket.

- **It parts by hand, twice per turnaround, with no tool**, which is what SVC-EEG-013 requires and what neither of the other two readings can do.
- **It does not change what is bought.** K2 stands as written; what changes is that its plug finally has something to enter. *That last clause is superseded on 2026-09-02. The K2 part does not change, but its purchase order gains two fields it never had – a stated plug-to-jaw lead length and a plug colour – for the reasons in 3.1.2.1 provisions 2 and 7. AVL-EEG-017 K1 already carries the identical correction one line above K2.*
- **It is touch-proof.** These two conductors reach a person’s ears through 47 kOhm from a +/-2.5 V rail. A bare crimped clip on 250 mm of free conductor is an accessible conductive part on a patient-applied lead, which is the thing DIN 42802 exists to prevent.
- **It is the one form of this part that is actually catalogued.** AVL-EEG-017 section 1.4.1’s whole difficulty with J15 to J17 is that the Staubli parts `design.py` names are “cable and panel parts, not PCB parts”. Here a cable part is exactly what is wanted, so this line does not carry the 12-week first-article risk the PCB socket line does.

- **And the guard.** The kit then has five 1.5 mm touch-proof interfaces – three EMG sockets on the pod panel and two ear sockets on the helmet temples – and an EMG lead will physically mate an ear socket. That is **not** a safety fault: all five sit behind their own 47 kOhm series resistor on the same +/-2.5 V rails, so the current bound of section 5.5 is unchanged. It is a measurement fault, and a silent one: an EMG lead in an ear socket puts a cheek electrode on the linked reference and the whole montage moves with it. **Test T10's lead-off check does not catch it**, because a mis-plugged electrode is still an electrode. The three defences are physical separation – panel against temple, 1500 mm apart – the colour code of section 10, and the placement guide of IFU-EEG-014. That is stated, not mitigated, and it is open item 24. **Corrected 2026-09-02: it is six interfaces, not five**, because section 3.1.3 adds the bias coupler at the halo front, and **all six are sockets**, so the count of ways to get it wrong is six plugs into six sockets in both directions. The count that matters is not how many but which half: with the instrument side all sockets, no helmet-mounted plug exists that could be carried into anything.

What changes in the wire list, and what does not. Conductors 9 and 10 keep their colours, their nets, their carrier paths, their "195 + 250 free" in-frame figure and their 2140 mm cut length: the socket replaces the crimp at the same place on the same tail, so no arithmetic in section 3.1 moves. What changes is the last row group of section 6 and the last 15 mm of the build.

Who has to decide. The **safety reviewer**, because this is a patient-applied touch-proof interface that is specified and not measured; **AVL-EEG-017**, because the coupler has no vendor and is carried there as OPEN WITH CRITERIA on line K27; and the **programme**, because the alternative – buying the clip bare and crimping it, which is what section 3.1 used to say – is a decision to accept a permanent joint and to rewrite SVC-EEG-013 R4, R5 and R10 around it. Until that is settled the two ear references cannot be built. That is open item 23.

3.1.2.1 The ruling of 2026-09-02, and the provisions that come with it

Ruling **D2-EAR-REFERENCE-COUPLER** keeps the clip and the coupler above and replaces the supporting provisions the proposal carried with the nine below, which have values. The clip is unchanged: **AVL-EEG-017 K2 exactly as written** – a Wuhan Greentek Ag/AgCl ear clip, finished, leaded, carrying its own DIN 42802 touch-proof plug. The coupler is unchanged: a free-hanging 1.5 mm touch-proof cable **socket to DIN 42802-1**, AVL-EEG-017 **K27**, crimped or soldered onto the 250 mm temple tail at the point the crimp used to be. **K27 is placed on the same purchase order as K2**, so that retention and finger-safety when the two are mated are one supplier's responsibility and not an argument between two of them; the Staubli LB-I1,5 cable-socket family is the approved alternate, and the exact order code, colour suffix included, is read off the catalogue on the day the order is raised and is not constructed here. **No cut length moves**: 2140 mm stands, as 1500 + 120 + 20 + 195 + 40 + 15 + 250.

1. The packing rule. This is the load-bearing provision and this document does not own the files it changes. The ear clips travel **mated and captive**, dressed inside the helmet bay. PKG-EEG-015 section 1.1 line 2.1 becomes `2 fitted | HELMET HM-01`; the (408, 162) 94 x 100 mm bay becomes EMG LEADS only; IFU-EEG-014 section 1's table and section 9 step 1 are re-issued; and SVC-EEG-013 R12 gains an explicit line that the ear clips are left mated and dressed inside the helmet bay, which closes the gap between its R10 and R12.

Why it is load-bearing. The proposal above argued that the mis-mate is an operator error, countable at R10. It is not, as the package stands. PKG-EEG-015 section 1.1 line 2.1 reads `Ag/AgCl ear-clip references | 2 | EAR CLIPS + EMG LEADS` – a bare "2", where every fitted line on that list reads "N fitted" – and section 2.2 puts HELMET HM-01 at (14, 14) in a 197 x 236 mm bay and EAR CLIPS + EMG LEADS at (408, 162) in a 94 x 100 mm one, with the three EMG DIN leads in the same hollow. IFU-EEG-014 section 1 renders that to the participant as "Two ear clips on a lead, and three coloured leads for the face pads", section 2 step 3 has the participant clipping them on, and section 9 step 1 sends them back to the same hollow. So without this provision the participant mates and unmates both couplers **every session, unsupervised, out of one 94 x 100 mm hollow holding five identical 1.5 mm DIN 42802 male plugs** – which is the exact fault the proposal claims to have removed – and IFU-EEG-014 section 8's "You dismantle nothing" is not true. With it, RISK-EEG-011 H-20's stated mitigation ("Participant touches nothing removable") becomes true rather than aspirational. **The programme lead signs this one**, because PKG-EEG-015 sections 1.1 and 2.2, IFU-EEG-014 sections 1 and 9, SVC-EEG-013 R12 and the RISK-EEG-011 H-20 and H-30 mitigations move together. It is open item 34, and it requires provision 2, or the clip cannot travel dressed inside the helmet bay.

2. The K2 lead length, and the deviation it extends. Apply the K1 correction verbatim to K2: the purchase order states the lead length and the termination. **Plug-to-jaw 150 to 200 mm** – a stated number, not the catalogue 1.0 to 1.5 m. The consequence has to be written down rather than absorbed: the unscreened reference run becomes the 250 mm temple tail plus at most 200 mm of clip lead, so **450 mm maximum, and it is recorded as an extension of DEV-WH-01 to REF_L and REF_R**. Section 3.6 accepts DEV-WH-01 for the three EMG channels only; section 4 argues at length why WH-01 is screened, and these two conductors are WH-01. **T8 – input-referred noise, 1.0 uV RMS maximum, an RTS-1 release criterion – is re-measured on a built unit carrying the real K2 lead before that extension is accepted.** If it fails, the clip lead is screened or the coupler moves to the earlobe end; length alone does not settle it.

3. Insulation of the free tail. A materials row in section 4, not a label citation: extruded PTFE or FEP sleeve, or 3:1 heat-shrink supplied at 1.5 mm or more and **recovered to 0.70 mm or less**, over the whole 250 mm free length, resisting 70 % IPA and the 40 kHz bath. *The Brady PS-187-2-WT class of section 10 is not an alternate here and no version of this section may cite it: it is a white thermal-transfer wire-marker sleeve at 1.6 mm recovered, which is more than twice the 0.70 mm OD of the conductor it would have to grip. A marker sleeve is identification, not insulation.*

4. Strain relief – NOT specified at this issue, and carried open. The proposal's 2.0 mm adhesive-lined bulb at a temple channel mouth, captured behind the cover strip, **is deleted**, because the released HM-01 has no temple channel mouth to put it in. `tools/mech_gen.py` defines exactly three mouths and all three are occipital: `HM01_N1_MOUTH` at (0.00, -95.38) in the shell roof, and `HM01_HALO_MOUTH` at (+/-45.41, -82.41, -11.25). `CH_BORE` is a constant 3.80 mm everywhere, so a 2.0 mm bulb has no shoulder to react against in any case. **Consequently test H6's 15 N leg is not applied to the ear couplers**, and the anchorage is carried open beside the identical OE-1 / OE-2 entry gap that KNOWN_ISSUES.txt already holds. That is open item 29. Until a temple wire exit exists in `tools/mech_gen.py` – a mouth stepped below 2.0 mm, or a moulded-in anchor – and is registered, there is nothing to pull against and nothing is tested.

5. Tests. Three changes, and one of them is a test the package has nowhere.

- **H3 is not extended to the couplers.** A DIN 42802 socket is single-pole and has no screen terminal (section 3.6 records the same fact as DEV-WH-01), step 6 cuts the WH-01 screen back at the helmet end so the tails are outside the screen, and H3's existing drain-to-every- conductor leg already covers what a coupler leg would have measured.
- **H4 gains them**: 500 V DC for 60 s, conductor bundle to any exposed metal **with the plug withdrawn**, with the socket body and its shroud counted as exposed metal, 100 MOhm minimum.
- **H8 gains them at 100 cycles minimum**, run as-built after step 15 with H1 repeated. The per-unit leg needs no new step: it lives in step 19's existing as-built repeat.
- **New H11, first article only**: an **IEC 60601-1 / IEC 61032 test-probe B** check on the unmated socket, recorded in the FAI pack. H4 cannot give this. H4 is a 500 V DC insulation-resistance measurement; finger-safety to the standard test finger is a gauge check, and until this issue the package substituted a supplier declaration for it.

6. The retention window, which K27 did not have. Separation force 5 to 15 N, stated and repeatable, added to K27 beside its existing mating-force and 500-cycle line. High enough that a snag does not part it – this is the one net whose partial loss section 3.1 itself calls "electrically almost invisible" – and low enough for SVC-EEG-013 R4's tool-free release. Every other separable interface in the kit already has a window (the bayonet at 3 to 6 N seating and about 10 N retention, H8's 0.15 N per contact, the boom at 30 N); this one had none.

7. Colour. GY on the left coupler and PK on the right stands, because it meets K27's actual criterion, "a colour distinct from the three EMG sockets". *The justification given as "the section 10 one-table rule" is withdrawn: that rule is about conductor identification, not about connector bodies.* Because K2 is bought unchanged, **the K2 purchase order gains a plug colour**: matched to the socket, or a printed GY / PK marker sleeve on the clip lead within 25 mm of the plug.

8. Life, and what it depends on. With provision 1 adopted, the couplers see roughly **100 operator mate cycles over five years** – 20 turnarounds per kit-year for five years – against K27's 500, a 5x margin, so the cycle count is not the binding criterion. If provision 1 is refused, the participant adds about **250** more (SVC-EEG-013 section 3: 25 sessions is about 10 turnarounds is about half a year, so roughly 50 sessions per kit-year), for about **350 – a 1.4x margin, not 5x**. Then three things change together and this ruling must be re-opened: K27's cycle criterion goes to **1000 minimum**; the colour code stops being an adequate defence against a participant choosing one of five identical plugs in one 94 x 100 mm bay, so the mis-mate has to be killed **mechanically**, by gender reversal or keying; and open item 24's residual is re-scored in RISK-EEG-011 as a per-session participant error rather than an operator error at R10. Gender reversal is not cost-neutral and is the second choice, not the first: the temple-tail plug form is a catalogue cable part, while a socket-ended ear clip is not a Greentek catalogue line and would be a bespoke Class A patient-contact electrode.

9. Open item 24, restated in both directions. An EMG lead will mate an ear socket, **and** the K2 clip's own plug will mate J15 to J17. With provision 1 both are operator errors at R10 and both are caught by R4's existing count-out of 8 cups and 2 ear clips. Without it, neither is.

Who signs this ruling. The **safety reviewer**, for the patient-applied touch-proof interface itself – K27 against its criteria including the new 5 to 15 N separation window, the H4 leg and the first-article H11 test-probe-B check on the unmated socket – and for accepting the extension of DEV-WH-01 to REF_L and REF_R **after** T8 has been re-measured on a built unit. The **programme lead**, for provision 1, because it is the change the whole safety argument rests on and it moves four other documents. The **mechanical reviewer with the PARTS / MECH owner**, for the temple wire exit, because a released model has to gain a feature and be re-cut before the H6 leg can be written at all. The **AVL-EEG-017 owner**, to place K27 on the same purchase order as K2 and to add the lead-length and plug-colour fields to K2. **ECO-EEG-016** carries the change record for all of it. **None of those signatures exists today.**

What would change it. A submitted K27 sample that is not finger-safe to IEC 61032 probe B with the plug withdrawn, or whose separation force is not repeatably inside 5 to 15 N over 500 cycles, kills the vendor and not the ruling – go to the Staubli LB-I1,5 alternate. If no maker supplies a cable socket that mates the K2 clip's plug with a stated retention force, the fallback is to buy clip and coupler as one assembled leaded pair, which makes the temple tail a permanent joint and forces SVC-EEG-013 R4, R5 and R10 to be rewritten around releasing at the clip jaw. A T8 above 1.0 uV RMS on a built unit means the unscreened reference run is not acceptable at any length.

3.1.3 WH-01 conductor 11 at Fpz, and WH-10 – the ruling of 2026-09-02

The bias pad is deleted as a helmet feature. *Superseded on 2026-09-02: the "Fpz bias pad, solder tag" of section 3.1's first issue, which was on no drawing, had no part number and appeared on no model. It was open item 26 and this section narrows that item rather than inventing the part.* Ruling **D3-BIAS-FPZ-TERMINATION** puts the electrode outside the helmet and the joint on the harness, for three reasons that stand without any of them being decisive alone: **there is no ninth cup, carrier, crown, spring, gel port or contact light anywhere in the package** for a ninth HM-04 to be made of; a disposable pad needs no retention feature and no reprocessing argument at all; and SVC-EEG-013 R5's 25-cycle disinfection compatibility protocol **has never been run**, so nothing reusable at that site can be qualified. *The 0.62 mm of section thickness that an earlier draft argued from – 12.55 mm needed against 11.93 mm available at Fpz – is withdrawn as a reason. It is real, but it is the smallest geometric deficit in the package: `mech_gen.py` records the same frame at 10.91 mm across the temple against the 12.20 mm this document's own channel section needs. An argument that would condemn the whole frame condemns nothing.*

What conductor 11 lands on. A **free-hanging 1.5 mm touch-proof socket to DIN 42802-1**, of the same **K27 class** as the two ear couplers – Staubli LB-I1,5 family or the Wuhan Greentek DIN 42802 cable range, both of which AVL-EEG-017 K27 already records as catalogued cable parts – in **TQ turquoise**, conductor 11's own colour and distinct from the GY and PK ear couplers and from the red / yellow / green EMG set. It hangs at the **HM-01 halo-front channel mouth**.

What mates it. WH-10 / WH-EEG-008-10, a new bought-in lead: **150 mm +/- 10 mm**, 4 mm female snap stud to a 1.5 mm DIN 42802 touch-proof **plug**, TQ turquoise, of the **K3 class** the three EMG leads are already bought in – reference Staubli SLS425-SEK/N, Greentek equivalent approved. AVL-EEG-017 **K47** buys it. This is the form suppliers actually build; a snap-to-socket lead is not a catalogue form, which is why the socket is on the harness and the plug is on the lead and not the other way round.

Gender, stated once. After this change **no plug exists anywhere on the instrument side**: three EMG sockets on the pod panel, two ear sockets on the temples, one bias socket at the halo front, and every mating half is a lead or a leaded electrode. A socket cannot be inserted into anything, so the driven output cannot be carried into another interface by hand.

The electrode. A fourth disposable pre-gelled Ag/AgCl snap pad off the existing K4 pack (Ambu BlueSensor N, 30 per pack), placed on prepped forehead skin **below the HM-02A brow pad**, one per session. **The kit therefore consumes four pads per session, not three**, so a 30-pack is seven full sessions with two pads left over. IFU-EEG-014 section 13.2's "Thirty are supplied, which is ten sessions' worth" is arithmetically wrong at four pads and is that document's to re-issue, together with its title "The three face pads", its "These are the only electrodes you position yourself" and its "Match colour to colour and you cannot get them the wrong way round". Whether a second pack is kitted is a programme decision and this document does not make it.

Cut length: 1940 + F, and F is not invented here. A free-hanging coupler needs a free tail; 1980 = 1500 + 120 + 20 + 285 + 40 + 15 has none, and its 40 mm site service loop has no site left to be coiled at. The cut is **1500 + 120 + 20 + 285 + 15 + F**, tolerance +10 / -0 mm on the fixed part. **F is stated explicitly on the wire list and is set at the first fitting trial**: bounded below by the clearance a hand needs to mate the coupler clear of the HM-02A brow pad, and above by the 250 mm the ear tails already carry. **Conductor 11's 1980 mm must not be carried forward into any cut schedule** – conductors 4 and 5 cut at 1980 mm for unrelated reasons and are unaffected. Open item 30.

Pull test: not H6's 15 N. H6's 15 N is defined as acting on HM-04A's anchorage in the body and on the solder joint; a free-hanging coupler has no body to anchor in, and 15 N is above the **13 N** minimum at which H5 qualifies a 28 AWG crimp. Either HM-01 gains a drawn anchorage that takes 15 N – which returns this to the mechanical reviewer with open items 22 and 26 – or **the coupler retention is set at or below H5's 13 N** and carries K27's mating-force and 500-cycle criteria instead. Open item 31. Nothing is pulled at 15 N here until the anchorage exists.

The cross-mate is stated, not designed out. Open item 24 grows from five interfaces to six sockets and six plugs, and it gains a consequence RISK-EEG-011 has not analysed: **a K2 ear clip or a K3 EMG lead will enter the bias socket**, putting an electrode on BIAS_EL, which is a driven output and not an input, at a site RISK-EEG-011 works only at Fpz. The reach is real and needs no lead: the halo ellipse is a = 81.10 mm lateral by b = 96.80 mm fore-aft (`mech_gen.py`), so temple to halo front is about 126 mm straight, and the ear tails are 250 mm free. If genuine non-interchangeability of the driven output is wanted, the answer is a different connector **class** or a keyed shroud, not gender – it is costable and it is referred to the safety reviewer, who is the only person who can decide that a colour is not enough.

This does not leave HM-01 untouched. A dressed channel mouth against 0.70 mm OD PTFE and a strain relief at the halo front are **new features on the carried-over HM-01 STL that no source file generates** (PARTS-EEG-019 OA-1: STL only, no STEP, no parametric source). The released frame has three channel mouths and all three are occipital, so the halo-front mouth this section needs does not exist any more than the temple mouths of 3.1.2.1 provision 4 do. **Three wire exits are now owed on one frame that cannot be regenerated from source.** Open item 26 is narrowed to that, and is not closed.

SR-12 stays open, and this change does not close it. Touch-proofing an interface does not alter what flows out through the electrode into the person. The number has moved and it has to be stated correctly rather than quoted from memory: with `tools/design.py`'s channel-11 override deleted on 2026-09-02, D11 and C11 sit on BIASOUT behind the full 47 kOhm like the other fifteen channels, so the shorted-clamp case collapses into the ordinary single-fault case of section 5.5 – **53.2 uA on bound A (2.5 V / 47 kOhm) and 41.2 uA on bound B (2.5 V / 60.615 kOhm) today, and 36.8 uA once ECO-EEG-024 raises the series resistors to 68 kOhm.** *The 183.6 uA that KNOWN_ISSUES and RISK-EEG-011 SR-12 record for this node, and the unbounded bound A beside it, are superseded by that change on 2026-09-02 and must not be re-quoted.* What is **not** superseded is that 53.2 uA is still over S-02's 50 uA single-fault limit and that **SR-12's disposition is the safety reviewer's to record**. RISK-EEG-011 offers three ways to close it – move D11 and C11 to the module side, add a second series element, or accept the topology with a written justification – and the first of those has now been done in the source. **None of the three is a connector.** No reviewer should read this section as closing SR-12.

Who signs. The **safety reviewer** first, on three things together: that a K27-class touch-proof socket is the correct patient-side form for the driven output; that the residual cross-mate above is **accepted and stated** rather than mitigated; and that SR-12 remains open. Then the **mechanical reviewer**, with open items 22 and 26, for the halo-front mouth, the dressed exit and the strain relief, and because the coupler retention value depends on whether an anchorage is drawn. Then **PARTS-EEG-019 and ECO-EEG-016**, to issue WH-10 / WH-EEG-008-10 – WH-01 to -07 and -09 are in use and WH-08 is withdrawn and not reused, so -10 is the next free number. Then **AVL-EEG-017**, to open K47 for the lead and to carry the bias socket as a third unit against K27's existing criteria. Then **IFU-EEG-014's owner**, for section 13.2. **None of those signatures exists today.**

3.1.4 The K12 preload spring – issued 2026-09-02, and why it can be issued now

Ruling D5-K12-SPRING-ENVELOPE declined to issue a dimensioned envelope, and it was right to, because the two things a vendor quotes against did not exist: the carrier had no rest position, since `hm04()` cut no circumferential bayonet run, and the only free volume above the spigot was 1.40 mm, which is a washer band and not a spring band. **Both facts changed on 2026-09-02 and both are measured, not asserted.** The run is cut – 100 degrees at r 4.30 to 5.55, z 1.10 to 3.80, over a 1.10 mm retaining lip – and `tools/simulate_production.py` measures 0.000 mm3 of interference through the quarter turn and through the carrier's 0.40 mm of axial travel. The seat roof moved from z 13.50 to **z 15.60**, so the free height above the spigot went from 1.40 mm to **3.50 mm**. The specification below is therefore derivable, and it is issued.

Superseded with it, on the same date and for the same reason: D5's provision that the part must be a multi-turn crest-to-crest wave spring, and its finding that a coil cannot be used. That finding was correct against a 1.40 mm band – the seat/spigot annulus is 0.10 mm on the radius, so no coil fits around the spigot and the spring must sit ON the spigot top, which made the free height the entire budget. With 3.50 mm of budget a coil fits, and a coil is what AVL-EEG-017 K12 has always said it was buying.

The geometry, read from `tools/mech_gen.py` at this issue *(model figures; no HM-04 and no HM-05B has been printed)*:

Datum	Value
HM-04 spring seat	6.80 dia, z 9.00 to 15.60
HM-04 gel port	2.50 dia through the seat roof, on the axis
HM-05B body / spigot	9.10 dia x 8.60 mm body in a 9.20 x 9.00 mm bore; 6.60 dia x 3.50 mm spigot; 2.40 dia passage on the axis
Spigot top, carrier fully down	z 12.10
Working height at rest	**3.50 mm** = 15.60 - 12.10
Working height at the hard stop	**3.10 mm**: the carrier body top meets the seat shoulder at z 9.00 after 0.40 mm of travel
Working stroke	**0.40 mm** – the travel the cup makes when it is pressed against the scalp, not slack

One discrepancy is reported and not resolved here. The lug sits at z 1.20 to 3.30 and the retaining lip roof is at z 1.10, so a carrier resting on its lip stands 0.10 mm proud of the cup face and its spigot top is at z 12.00, which makes the rest height **3.60 mm** and the stroke **0.50 mm**. The 3.50 / 0.40 figures above are the ones `mech_gen.py` states, taken from the 9.00 mm bore against the 8.60 mm body. The two readings differ by one printed layer and the difference belongs to `tools/mech_gen.py`, which this document does not own – open item 33. **The spring below is specified so that it does not matter**: it holds the force window over every installed height from 3.10 to 3.60 mm, and over the tolerance band around both.

The tolerance band the spring must work over. MJF is +0.15 / -0.05 mm and the package gates that fit at 9.20 / 9.35 / 9.15 mm on FIT-01. The installed height is a stack of two printed dimensions – the HM-04 seat depth and the HM-05B body-plus-spigot height – so **+/- 0.25 mm** is allowed on it (+/- 0.30 worst case, +/- 0.21 root-sum-square, rounded up). The spring must therefore hold its window over installed heights of **2.85 to 3.85 mm**.

The specification, and the arithmetic behind every number. Force target 3 to 6 N is AVL-EEG-017 K12's own stated target and is **not** derived here; everything else is derived from it and from the geometry above.

Characteristic	Value	Where it comes from
Form	helical compression, **closed and ground both ends**	the 3.50 mm band now takes a coil; a ground end is required so the load spreads over a full turn of PA12 rim rather than a short arc
Material	**stainless AISI 302 spring temper to ASTM A313**, as K12 says	K12; see the open material question below

Characteristic	Value	Where it comes from
Wire diameter d	**0.50 mm**	the largest wire that reaches 1.20 N/mm inside the 6.50 / 5.40 mm envelope and still stacks short: at 0.55 mm the same rate needs 3.1 active coils and a solid height of 2.82 mm, over the 2.60 mm limit below
Outside diameter	**6.40 mm nominal, 6.50 mm maximum** free and at every point of the stroke	the seat bore is 6.80 mm at +0.15 / -0.05, so 6.75 mm worst case: 0.25 mm of diametral clearance at the tight end. Calculated coil growth at solid is 6.46 mm, so the 6.50 ceiling holds through the stroke
Inside diameter	**5.40 mm minimum**	so the coil bears on the PA12 spigot rim between r 2.70 and r 3.20, inside the rim's r 1.20 to 3.30 and clear of the 2.40 mm passage; at the top it bears between the 2.50 mm gel port (r 1.25) and the bore wall (r 3.40). **Neither end covers the gel port**
Mean coil diameter	5.90 mm; spring index 11.8	6.40 - 0.50
Active coils	**2.2**	k = G d^4 / (8 D^3 n) with G = 69 000 MPa for 302: 69 000 x 0.50^4 = 4 312.5, 8 x 5.90^3 = 1 643.0, so k = 2.625 / n and n = 2.19 for 1.20 N/mm
Rate k	**1.20 N/mm +/- 10 %**	chosen low on purpose: the whole 3 to 6 N window is 3 N wide, and 0.50 mm of stroke plus +/- 0.25 mm of build tolerance is 1.00 mm of height variation, which at 1.20 N/mm is 1.20 N of it
Free length	**6.90 mm +/- 0.20 mm**	set so the worst-case low still preloads: 1.08 N/mm x (6.70 - 3.85) = **3.08 N**, above the 3 N floor
Solid height	**2.10 mm calculated; 2.60 mm maximum**	4.2 total coils x 0.50 mm. The maximum is the ruling's own relation, solid height at least 0.25 mm below the minimum working height of 2.85 mm
Squareness	3 degrees maximum	it is unguided at both ends

Characteristic	Value	Where it comes from
Ends	ground square, deburred	PA12 bearing

The forces that result, calculated:

Condition	Installed height	Force
At rest, nominal build	3.50 mm	1.20 x (6.90 - 3.50) = **4.08 N**
At rest, on the lip reading	3.60 mm	**3.96 N**
At the hard stop, nominal	3.10 mm	**4.56 N**
Worst case low, k and L0 both low, build tall	3.85 mm	1.08 x (6.70 - 3.85) = **3.08 N**
Worst case high, k and L0 both high, build short	2.85 mm	1.32 x (7.10 - 2.85) = **5.61 N**
If it were stacked solid anyway	2.10 mm	1.20 x 4.80 = **5.76 N**, and about 6.7 N at the worst case of rate, free length and wire tolerance together

Three consequences worth stating, because they are what the low rate buys.

- **The force is bounded by the spring and not by the assembler.** The carrier reaches its mechanical stop – body shoulder on seat shoulder – with 1.00 mm of spring travel still in hand at nominal and 0.75 mm at worst case, so the spring cannot stack solid inside the stroke. And even if it did, it delivers under 6.7 N. The failure the ruling was most concerned by – a rigid metal stack handing the scalp whatever the helmet presses with, for a two-hour session, on a patient-applied part – cannot happen at this rate.
- **The PA12 rim is not overloaded.** A ground end bears over about one full turn, pi x 5.90 x 0.50 = 9.3 mm2, so 4.56 N is **0.49 MPa** against PA12's compressive strength of tens of MPa. An unground end would land on a short arc and multiply that by roughly three, which is why the ground end is a requirement and not a preference.
- **The steel is not near its limit.** Wahl factor 1.12 at index 11.8; shear stress 8 F D Kw / (pi d^3) is **615 MPa** at the 4.56 N working maximum, 809 MPa at the 6 N ceiling and about 900 MPa if it were stacked solid at worst case, against a minimum tensile of about 2 000 MPa for 0.50 mm 302 spring wire – 31 %, 40 % and 45 %. Helix angle 8.2 degrees, under the 12 degrees at which the rate formula stops holding. Free length over mean diameter is 1.17, well under the 2.6 at which a squared-and-ground spring buckles between parallel faces.

The no-gold relation, and it does not close at the crown diameter this document proposes. The spring must not touch HM-05C: not because a stainless ring bearing on gold is a second metal junction in parallel with the measurement – seated on PA12 at one end, it is a dead-end stub – but because of **galling of the gold-over-nickel plating K25 and K26 require, under the 90-degree turn SVC-EEG-013 R4 performs at eight sites every turnaround**. The rule has to be a relation and not two independent limits, because float is what defeats it:

(ID minimum - crown OD maximum) / 2 >= (seat bore maximum - spring OD minimum) / 2 + 0.05 mm

With ID 5.40 minimum, spring OD 6.30 minimum and a seat bore of 6.95 mm at the loose end of MJF, the right-hand side is 0.375 mm and the relation requires a **crown outside diameter of 4.65 mm or less**. Section 3.1.1's crown is proposed at **5.20 mm**, which fails it by 0.55 mm on the diameter: at the extreme of its float the coil reaches the gold. Three ways out, and the choice is the **mechanical reviewer's on MECH-EEG-020 sheet 8**, not this document's and not a purchase specification's:

1. bring the crown outside diameter down to 4.65 mm or less;
2. recess the crown's upper face **0.15 mm below the spigot rim** – equivalently, stand the rim 0.15 mm proud of the crown – so that an overhanging coil cannot reach it. HM-04A's leaf then travels 0.15 mm further, which it is sprung to do;
3. locate the spring radially to +/- 0.10 mm with a feature on the rim or in the seat.

The spring is not part of the electrical joint, and that is a measurement and not a claim. It bears on printed PA12 at both ends. The first-article check below measures it.

Qualification regime. 25 cycles of **SVC-EEG-013 R6** – two 10 ml warm-water passes, demineralised rinse, 1 bar air or less – plus a 70 % IPA wipe and a conductive-paste dwell: the same regime K25 carries. **Not the cups' 40 kHz R5 bath.** ASM-EEG-007 section 4.2 bonds HM-04 into HM-01 and SVC-EEG-013 R5 lists HM-01 as wipe, never immersed, so this spring never sees the bath even though the cup it sits above does.

What must be measured on the first five, before any fleet order. K12 already requires five pieces; these are what is measured on them, **on a printed HM-04 / HM-05B pair and not only on the bench**:

#	Measurement	Accept
1	Free length, each piece	6.90 +/- 0.20 mm
2	Solid height, each piece	2.60 mm maximum
3	Rate, from force at 3.60 mm and at 3.10 mm	1.20 N/mm +/- 10 %
4	Force at 3.60 mm and at 3.10 mm	inside 3.0 to 6.0 N at every height from 2.85 to 3.85 mm
5	Outside diameter free and compressed to 2.85 mm; squareness	6.50 mm maximum; 3 degrees maximum
6	Seating in the printed pair, spring pushed to one side of the seat	the coil bears wholly on the spigot rim and **does not touch HM-05C** – the go / no-go against the no-gold relation
7	Gap between the carrier's mechanical stop and the spring's solid height	0.25 mm minimum, measured, not calculated

#	Measurement	Accept
8	Resistance, spring to HM-05C and spring to HM-04A, dry; and H1 site-to-connector with the spring removed	10 MOhm minimum; H1 unchanged. **This is the check that the preload member is not the conductor**
9	After 25 cycles of the R6 + IPA + paste regime: free length, force at 3.60 mm, and a 20x inspection under the coil ends	free-length loss 5 % (0.35 mm) maximum; force still 3.0 N minimum; no crevice pitting
10	With the spring fitted: the bayonet still turns with the HM-09 key, and SVC-EEG-013 R10's 10 N straight pull still does not release the carrier	as R10

The material question the ruling raised and this specification does not settle. K12 says 302 and 302 is what is specified above, because that is the line the programme is buying against. The ruling's finding stands beside it and is not dismissed: **the seat is a chloride crevice under the coil ends that is never immersed, never scrubbed and never fully dried**, and on that argument 316 spring temper would be mandatory rather than an alternate. Measurement 9 is what decides it. If it shows pitting or relaxation, the line goes to **316**, and beyond that to Elgiloy; the change costs a purchase-order line and no geometry, which is why it is safe to resolve it on evidence rather than on argument.

Still absent, and named so that nobody reads this section as more finished than it is. The spring has **no retention or capture feature**: nothing holds it if a carrier is drawn without one. **ASM-EEG-007 has no step that fits it and SVC-EEG-013 has no step that handles it** (KNOWN_ISSUES.txt records that no assembly step fits the cups, the carriers or the springs at all). Section 8 step 14a of this document is the harness half of that and it is not enough on its own. Open item 32.

Who signs, and the order matters. (1) The **mechanical reviewer on MECH-EEG-020 sheet 8**: the crown dimension, so the no-gold relation becomes a pair of limits rather than a failure; a capture feature for the spring; the one-time allocation of the z 11.70 to 14.30 band across HM-04A's anchorage, the LED seat and its two lead passages and the dressed conductor exit; and the 0.10 mm rest-datum question of open item 33. (2) The **safety reviewer**: this is a patient-applied part, the solid-height bound is what keeps the force bounded, and a stainless preload member inside a gel-flushed volume is signed as an accepted open item or designed out. (3) **AVL-EEG-017 with a spring maker** – K12 names Lee Spring, Century Spring and Gutekunst – against the five measured samples above, and only after (1). (4) **ECO-EEG-016**, to carry the change to this section and to AVL-EEG-017 K12, K25 and K26, because this contradicts a live proposal rather than silently overwriting it. **SAMPLE ONLY until all ten measurements pass. No fleet order.**

3.2 WH-02 – helmet contact-light cable, 10-way, to J30

Insulation PVC. Same 7/0.1 mm tinned-copper conductor. Cut lengths use the same 1640 mm pre-frame allowance as WH-01, then the channel-B routed length + 55 mm.

Cond	Colour	Gauge	Insul.	From (helmet terminal)	To (pin)	Net	Carrier path	In-frame (mm)	Cut (mm)
1	WH/BK	28 AWG eq.	PVC	Fz LED, lead A	J30.1	LED1	R70 1k -> SR_Q0 (J19.8)	330	2025
2	RD/BK	28 AWG eq.	PVC	Cz LED, lead A	J30.2	LED2	R71 -> SR_Q1 (J19.9)	220	1915
3	BU/BK	28 AWG eq.	PVC	Pz LED, lead A	J30.3	LED3	R72 -> SR_Q2 (J19.10)	145	1840
4	GN/BK	28 AWG eq.	PVC	C3 LED, lead A	J30.4	LED4	R73 -> SR_Q3 (J19.11)	300	1995
5	YE/BK	28 AWG eq.	PVC	C4 LED, lead A	J30.5	LED5	R74 -> SR_Q4 (J19.12)	300	1995
6	BN/BK	28 AWG eq.	PVC	T7 LED, lead A	J30.6	LED6	R75 -> SR_Q5 (J19.13)	190	1885
7	OG/BK	28 AWG eq.	PVC	T8 LED, lead A	J30.7	LED7	R76 -> SR_Q6 (J19.14)	190	1885
8	VT/BK	28 AWG eq.	PVC	F7 LED, lead A	J30.8	LED8	R77 -> SR_Q7 (J19.15)	265	1960
9	GY/BK	28 AWG eq.	PVC	WH-BUS-01 pad 9	J30.9	LED_V	R78 (0R) from LED_PWM, GPIO48	30	1725
10	BK black	28 AWG eq.	PVC	WH-BUS-01 pad 10 (isolated)	J30.10	LED_GND	R79 (0R) to DGND	30	1725

Eight LED_V tails leave WH-BUS-01 and run beside their matching LEDn conductor to the second lead of each site LED:

Tail	Colour	From	To	Length
WH-02T-1 to -8	GY/BK	WH-BUS-01 pads 1 to 8	Fz, Cz, Pz, C3, C4, T7, T8, F7 LED lead B	matching LEDn in-frame length less 30 mm

LED_GND carries no LED current in either drive phase. The contact lights are two-lead bicolour devices between LEDn and the LED_V common: phase A drives LED_V high and Qn low (green), phase B drives LED_V low and Qn high (red), and alternating fast enough reads as amber. The requirement is that the alternation is above 100 Hz; the fitted value is **240 Hz**, `LIGHT_PHASE_HZ` in `board_pins.h`. The return in both phases is LED_V. LED_GND is a 0 V guard conductor, held at DGND through R79, laid in the geometric centre of the WH-02 bundle so that the eight switched lines always have a 0 V neighbour and their displacement current returns beside its source rather than through the wearer. It is landed on an isolated pad of WH-BUS-01 and connected to nothing else. It exists so the return topology can be changed by ECO without a new cable.

The bicolour phase scheme is specified and not yet coded. `lights_write()` and `lights_task()` in the firmware are on and off only, per FW-EEG-001 section 1.1, so the amber state that TST-EEG-004 T11 tests cannot pass until the driver is written. The cable is built for the scheme regardless, because the wiring is the same either way.

Drive current is (3.3 - 2.0) / 1000 = **1.3 mA per site**, so **10.4 mA total** sourced or sunk by GPIO48 with all eight lit. LED_V is GPIO48, an input at reset, so nothing on the head can light at boot whatever the shift register happens to contain, and all eight lights are forced dark during recording blocks (RFQ E-27).

3.2.1 WH-BUS-01 – the contact-light bus board

WH-BUS-01 is what makes the sentence “there are no splices anywhere in the kit” true. LED_V arrives at N1 as one conductor and has to become nine – the incoming conductor plus eight tails – and the alternative to a board is eight crimp splices buried under a cover strip inside a helmet that a participant puts on and takes off. PARTS-EEG-019 Rev B registered the part, gave it a size and a pad count, and recorded that no Gerber set existed for it. That set now exists: `kicad/wh-bus-01/`, written by `tools/wh_bus.py`, with a README beside it that carries the stack-up, the finish and the panel.

Property	Value
Size	**14.0 x 10.0 mm**, rectangular, no cut-outs, no slots, no mounting hole
Layers	**two**, both signal. No plane and no pour: there are two nets on this board
Stack-up	mask / 35 um L1 / FR-4 core 0.71 / 35 um L2 / mask = **0.80 mm +/- 10 %** finished
Material	FR-4, Tg >= 150 C, 1 oz (35 um) copper both sides – the carrier’s material, so one qualification covers both boards
Finish	ENIG, Au 0.05-0.10 um over Ni 3.0-6.0 um, as EEG-CAR-01. Lead-free HASL is acceptable on notice
Mask / legend	green LPI both sides; white legend both sides. The bottom legend is mirrored in the data
Pads	ten, 1.60 mm, on a 2.70 x 4.80 mm grid, two rows of five. Pad 9 is square
Plated holes	ten, **0.80 mm finished**, one drill tool, 0.40 mm annular ring
Minimum clearance used	1.10 mm, LED_V to the LED_GND island. Nothing here is near a fabricator’s limit
Panel	5 x 4 V-scored array, 20 up, step 14.0 x 10.0 mm, 5 mm rails: an 80.0 x 50.0 mm panel
Electrical test	100 % to the supplied IPC-D-356A netlist

Two layers, where the register says one. The pads are plated through holes, and plating needs two layers. A 28 AWG 7/0.1 mm conductor soldered to a surface-only pad is held by the pad’s peel strength alone; through the board and soldered on both sides it is held by the barrel. The board sits in a helmet under a cover strip and is handled every time a conductor is replaced, so that difference is the whole argument. The second layer also doubles the bus copper and costs nothing at these quantities. **PARTS-EEG-019 Rev B’s “single-layer FR-4” wording and its “no Gerber set has been generated for it” status were both superseded by this data set, and PARTS-EEG-019 is corrected on both counts at its 2026-09-02 issue**; that closes open item 15.

The hole size is derived, not measured. 1.60 mm is the registered pad. 0.80 mm is the hole that pad wants for the conductor it takes: the WH-02 conductor is 7/0.1 mm tinned copper, about 0.30 mm across the strand bundle (section 4), and 0.80 mm leaves 0.25 mm all round for tinning and for solder to wick. It is not the carrier's 1.00 mm socket-strip hole; the two boards share no drill programme, and 1.00 mm would only mean more solder in each joint. No joint has been made yet, so this is arithmetic, not experience.

No resistor goes on this board. The eight series resistors are R70 to R77 on the carrier, 1 kOhm each, between the 74HC595 outputs at J19.8-15 and the cable, and the common reaches the cable through R78, 0 ohm. The 1.3 mA per site is already set there. A resistor here would be a second limit in the same loop and would change a current that this document, ICD-EEG-006 and TST-EEG-004 all quote. Ten pads, one copper bar, nothing fitted – so there is no CPL file, no paste layer and no stencil for this part, and their absence is the design rather than an omission.

Pad map. Fabrication coordinates, origin at the bottom-left corner with Y up, the same convention as the Gerbers, the drill file and the netlist.

Pad	Net	x, y (mm)	Shape	What lands on it
1	LED_V	4.50, 7.40	round	tail WH-02T-1, to the Fz LED lead B
2	LED_V	7.20, 7.40	round	tail WH-02T-2, to the Cz LED lead B
3	LED_V	9.90, 7.40	round	tail WH-02T-3, to the Pz LED lead B
4	LED_V	12.60, 7.40	round	tail WH-02T-4, to the C3 LED lead B
5	LED_V	4.50, 2.60	round	tail WH-02T-5, to the C4 LED lead B
6	LED_V	7.20, 2.60	round	tail WH-02T-6, to the T7 LED lead B
7	LED_V	9.90, 2.60	round	tail WH-02T-7, to the T8 LED lead B
8	LED_V	12.60, 2.60	round	tail WH-02T-8, to the F7 LED lead B
9	LED_V	1.80, 7.40	**square**	WH-02 conductor 9, GY/BK, LED_V from J30.9
10	LED_GND	1.80, 2.60	round	WH-02 conductor 10, BK, LED_GND from J30.10

Pad 9 is square, and that is the only orientation feature the board has: the square pad and the "9" on the legend mark the input end, which faces OE-2. Fitted the other way round the eight tails still reach their LEDs, but the input pair has to cross the whole board and the N1 cover strip will not close over the crossing conductors.

Pad 10 is an island. It is 1.10 mm from the nearest LED_V copper on both layers and it is connected to nothing, which is the point: LED_GND is a 0 V guard conductor and must not become a return path (open item 9). That isolation is also the one property of this board that a visual inspection will not catch, which is why the IPC-D-356A netlist ships with the Gerbers and why the electrical test is not optional. If an ECO ever ties LED_GND to something, the tie is a wire link off pad 10, not a new board.

What is still unknown. No WH-BUS-01 has been fabricated, soldered or measured; the clearances, the annular ring and the copper-to-edge figures above are computed by `tools/wh_bus.py --check` from the geometry that produced the Gerbers, and nothing else about the board has been verified. V-scoring 0.80 mm material is at or near the lower limit at several fabrication houses, so the panel line is the one most likely to come back as a question; the README states the tab-routed alternative. And **HM-01 has no pocket, no boss and no location feature for this board at N1** – it is retained by the cover strip and by its own ten solder joints, and no drawing says where within N1 it beds. That is open item 16.

3.3 WH-03 – boom microphone pigtail, 4-way, to J18

From J18 on the carrier to the 4-conductor 3.5 mm panel jack on adapter WH-ADP-01, which is specified in section 3.9. Length 220 mm. Rev A put that jack “3 mm behind a recessed 7.0 mm opening in the POD-P1 underside”; `tools/mech_gen.py` cuts a **6.5 mm opening in the right-hand wall** at design (122.0, 90.0) and cuts nothing in the underside, so the jack face finishes 2.5 mm behind the outer face – the wall thickness – and the Rev A sentence is withdrawn.

Cond	Colour	Gauge	Insul.	From	To	Net	Note
1	RD red	28 AWG eq.	PTFE	J18.1	jack tip (T)	VOICE_RAW	high-impedance electret line, on to J21.4, the preamplifier input on MP-01
2	BK black	28 AWG eq.	PTFE	J18.2	jack ring 1 (R1)	DGND	capsule return
3	BK black	28 AWG eq.	PTFE	J18.3	jack ring 2 (R2)	DGND	screen return at the jack
4	BK black	28 AWG eq.	PTFE	J18.4	jack sleeve (S)	DGND	shell
–	bare drain	30 AWG	–	J18.2 (pod end only)	cut back at the jack	DGND	see section 5.3

The preamplifier is on MP-01, not on the boom and not on the carrier. The boom carries the bare electret capsule and its screen, and nothing else. VOICE_RAW arrives on this pigtail at J18.1, crosses to J21.4, and the amplified VOICE_PRE returns on J21.3 for the codec at J9.1 and for envelope channel 2. `design.py` governs: J21 is a carrier socket, and purchased modules sit on MP-01 and connect through keyed jumpers, so the module in the J21 jumper is physically on the plate. This settles the v1 conflict in which DESIGN_FACTS section 2, RFQ section 2, RFQ E-14 and DSN-EEG-003 section 2 all put the preamplifier on the boom, and Rev A of this document put it on the carrier. All of those are corrected to MP-01. Keeping the amplifier off the head keeps 3.3 V off the head, keeps the boom immersible in the ultrasonic refurbishment bath, and costs a high-impedance run that section 4 handles by screening it and keeping it short.

Which preamplifier is not settled, and this document does not settle it. The MAX9814 named throughout package v1 is an automatic-gain-control part, and RFQ E-14 requires AGC off; disabling it is a module-dependent modification. The preferred route is a fixed-gain part of the MAX4466 class, such as Adafruit 1063. Until a part is bought and measured the module is specified **by interface only**, in ICD-EEG-006, and AVL-EEG-017 carries the MAX9814 as **not approved**. Nothing on this pigtail changes with the choice; what changes is R89 (2k2, DO NOT POPULATE), the electret bias, which is fitted only if the module finally chosen does not supply its own, per ICD-EEG-006 section 7.2.

MIC_MUTE does not reach the boom in Rev B. The mute line exists only at J9.4 and J28.4, both inside the pod. No logic edge travels to the head.

WH-03B, the boom lead itself, is 1700 mm: capsule + to tip, capsule - to ring 1, screen to ring 2 and sleeve, terminated in a moulded 3.5 mm 4-conductor plug.

3.4 WH-04 – headphone panel pigtail, 4-way, to J27

From J27 to a second 4-conductor panel jack of the same part on adapter WH-ADP-01B, which is specified with WH-ADP-01 in section 3.9. Length 190 mm. Using the 4-pole part for a stereo output is deliberate: the fourth contact becomes the insertion detect that J27.4 reserves, and the kit then carries one jack part number, not two.

Cond	Colour	Gauge	Insul.	From	To	Net
1	WH white	28 AWG eq.	PTFE	J27.1	jack tip	HP_L
2	RD red	28 AWG eq.	PTFE	J27.2	jack ring 1	HP_R
3	BK black	28 AWG eq.	PTFE	J27.3	jack sleeve	HP_GND
4	GY grey	28 AWG eq.	PTFE	J27.4	jack detect contact	NC_HP_DET

HP_TAP, the stimulus envelope source, is taken on the codec module at J8.10 and never appears on this cable, which satisfies RFQ E-16’s requirement that the tap sit before any user-adjustable control.

The headphones this pigtail drives are specified in RFQ A-04 as **32 to 64 Ohm**, and the calibrated output level is measured per model; the shipped ATH-M20x is 47 Ohm. The harness is indifferent to the impedance, but the acoustic limit is not: RFQ E-29 caps the output at **100 dB SPL** at any commanded level, against a calculated full-scale output of about 110 dB SPL, and that clamp lives in the firmware volume register, not in this cable. The requirement is therefore **not met by hardware here** and cannot be, and this pigtail does nothing to enforce it.

3.5 WH-05 – room microphone cable, 4-way, to J28

From J28 to the room-microphone module mounted behind the meshed acoustic port in the POD-P1 wall. Length 180 mm.

Cond	Colour	Gauge	Insul.	From	To	Net
1	RD red	28 AWG eq.	PTFE	J28.1	module VDD	DVDD3V3
2	BK black	28 AWG eq.	PTFE	J28.2	module GND	DGND
3	WH white	28 AWG eq.	PTFE	J28.3	module OUT	ROOM_PRE
4	YE yellow	28 AWG eq.	PTFE	J28.4	module MUTE	MIC_MUTE

Cond	Colour	Gauge	Insul.	From	To	Net
–	bare drain	30 AWG	–	J28.2, pod end only	cut back at the module	DGND

The mute must be a hardware gate in the module's signal path, driven by GPIO21, and not a firmware-only mute, which is what RFQ E-15 asks for. **No room-microphone module is known to meet E-15**, so the interface above is specified against a candidate and not a qualified part: a Knowles SPU0414HR5H-SB class analogue MEMS capsule with a TI TS5A3159 analogue switch in the output path, carried on adapter WH-ADP-02, which is specified in section 3.9, with the module interface governed by ICD-EEG-006. If no catalogue module qualifies, the fallback is that same capsule and switch as a programme-designed sub-assembly, which needs a drawing, a part number and an AVL line that none of them has today. This wire list does not change in either case.

3.6 WH-06 – EMG DIN lead set, 3 off

Bought-in, not built: 1000 mm lead, 4 mm female snap stud to a touch-proof 1.5 mm DIN 42802 plug. Reference part Staubli SLS425-SEK/N, 1.0 m; approved alternate Greentek equivalent.

Lead	Colour	Site	Socket	Net	Carrier path
WH-06-1	RD red	cheek	J15	EMGIN1	R12 -> D12 -> C12 -> EMG1 (J4.1)
WH-06-2	YE yellow	submental (chin)	J16	EMGIN2	R13 -> D13 -> C13 -> EMG2 (J4.2)
WH-06-3	GN green	laryngeal (neck)	J17	EMGIN3	R14 -> D14 -> C14 -> EMG3 (J4.3)

The lead colours are red, yellow and green because that is the ruling, not because this document prefers them. Rev B as first issued bought these three leads as white, brown and grey while IFU-EEG-014 section 13.2 and PKG-EEG-015 section 4.2 named red, yellow and green for the same three sites, and this section reported that as undecided. It is decided. **IFU-EEG-014 Rev B section 13.2 rules the code red / yellow / green**, in its control note, and carries the ruling in its section 16 item 11; the reasoning recorded there is that the person who has to resolve a colour difference is the participant, matching a lead in the hand against a coloured socket legend under a desk lamp, and white against grey is not a distinction to ask of anyone in a mirror. The code the participant reads is the code that governs, so this document is re-issued to it rather than the other way round.

Nothing in this document depended on the white / brown / grey names: the three leads are distinguished by socket and by net everywhere else, and WH-06 is bought in rather than built. IFU-EEG-014 records that the lead set is a catalogue item stocked in either code, so the ruling costs nothing at the harness and nothing in lead time. What it does do is make the colour a controlled characteristic: the Staubli SLS425-SEK/N line and any approved alternate must name red, yellow and green on the order, and goods-in checks the three colours against this table before the set is kitted. Former open item 7 of section 11 is closed by this correction and its number is not reused.

The **carrier sockets at J15 to J17 have no confirmed part.** `design.py` names Staubli SLB1,5-F / LB-I1,5, but those are cable and panel parts, not PCB parts, and no catalogue part has been confirmed to fit the footprint: a touch-proof 1.5 mm socket with a PCB-mount signal pin and two 1.5 mm retention posts. It must be sourced and first-articled before Phase 2 and AVL-EEG-017 carries a 12-week lead-time risk against it. What the footprint requires is fixed: a 1.70 mm plated hole for the signal pin with six 1.50 mm non-plated retention posts that carry no copper and no mask (ECO-EEG-012). This is a Class A patient-contact part on every unit and it has no vendor.

Deviation DEV-WH-01. DSN-EEG-002 Rev E section 6 calls the EMG runs “3 x screened”. A DIN 42802 touch-proof socket is single-pole and has no screen terminal, so there is nowhere for a screen to land. The EMG leads are unscreened. This is recorded as a deliberate deviation rather than a defect; the alternative – a ground stud beside the panel wired to AGND_REF – was rejected because it puts an exposed conductor referenced to the analogue mid-rail on the outside of the pod. DSN-EEG-002 is corrected at its next revision.

3.7 WH-07 – charge-port pigtail, 2-way, to J24

From J24 (JST B2B-PH-K, 2.00 mm pitch) to the charge-only USB-C receptacle on adapter WH-ADP-03, which is specified in section 3.9. Length 150 mm.

Cond	Colour	Gauge	Insul.	From	To	Net
1	RD red	24 AWG	PVC	J24.1	receptacle VBUS (A4, A9, B4, B9 commoned)	VBUS_IN
2	BK black	24 AWG	PVC	J24.2	receptacle GND (A1, A12, B1, B12 commoned)	DGND

24 AWG here, not 28, because this pair carries the full charge current: F1 is a 1.1 A hold / 2.2 A trip PTC and the conductor must not be the fuse. Calculated loop resistance at 0.079 Ohm/m for 24 AWG over 0.3 m is 0.024 Ohm; at 1.1 A that is 26 mV, which is inside the charger’s input tolerance.

The receptacle carries two 5.1 kOhm CC pull-downs so a Type-C source presents 5 V. **No data conductor enters this cable.** D+, D-, SBU and the CC lines terminate on the adapter. The receptacle shell is isolated from DGND – see the open item in section 11.

Charging and recording are mutually exclusive by the two independent mechanisms of RFQ S-01, which are written out once in DESIGN_FACTS section 4 and are not restated here. The consequence for the harness builder is the one that matters: **the helmet is never worn while this cable is connected**, so WH-07 and WH-01 are never live at the same time.

The temperature half of that interlock is **not met**. RFQ E-23 asks for a charger with thermal regulation and no charging above 45 C, and RFQ S-04 asks for a thermistor; there is no NTC net in `design.py` and no thermistor way on J12 or J13, so no conductor in this document carries a cell temperature and none can be added without a board change. It is an open hardware item listed in DSN-EEG-003 section 11 and RISK-EEG-011.

3.8 WH-09 – isolator host pigtail, USB-B to USB-C panel receptacle

This assembly replaces the withdrawn WH-08 and exists to resolve a connector mismatch, so the mismatch is stated first.

RFQ E-24 asks for a USB-C host connector. The only named candidate isolator module, the Olimex USB-ISO class ADuM4160 board, presents a USB-B host receptacle. That is a live non-conformance and it is not closed. The interim resolution, which is what is built for Phase 1, is this short pigtail: a moulded USB-B plug at the module, a panel-mount USB-C receptacle at the POD-P1 aperture. If and when an isolator module with a USB-C host receptacle is qualified, that module's own receptacle is presented directly through the aperture, WH-09 is deleted, and E-24 is met by the module rather than by a pigtail. Until then the kit meets E-24 at the panel and not at the module, which is a difference a safety and EMC reviewer must see rather than have hidden by a cable.

Length 150 mm. Reference construction: a shielded USB 2.0 cable assembly, 28 AWG data pair, 24 AWG power pair, OD 4.0 to 4.5 mm, with the panel receptacle on adapter WH-ADP-04, which is specified in section 3.9.

Conductor	From	To	Note
VBUS	panel receptacle A4/A9/B4/B9	module USB-B pin 1	powers the host side of the isolator only; no path to the battery rail
D+	panel receptacle A6/B6	module USB-B pin 3	
D-	panel receptacle A7/B7	module USB-B pin 2	
GND	panel receptacle A1/A12/B1/B12	module USB-B pin 4	host-side 0 V, not DGND
braid	panel receptacle shell	module USB-B shell	bonded at both ends of this cable and **to nothing else**; not bonded to DGND anywhere in the pod

The panel receptacle carries two 5.1 kOhm CC pull-downs, so a Type-C host enumerates the pod as a device and supplies 5 V. Every conductor in this assembly is on the **host** side of the isolation barrier. Nothing in it may be commoned with DGND, with the WH-07 charge return, or with any harness drain, because that is the one bond the ADuM4160 exists to prevent.

No carrier copper crosses the isolation barrier (RFQ E-24, S-03). The keep-out that guarantees it is specified once, in DSN-EEG-003 section 3.3, and this document does not restate its coordinates; what the harness builder must know is that it is a keep-out on **all four layers**, not two, and that nothing in this assembly, including a cable tie, a P-clip screw or a stray drain, may be routed across it.

Retention and sealing. The panel receptacle is fixed to the POD-P1 wall on adapter WH-ADP-04 with a gasket behind its flange; the aperture is a gasketed panel aperture and **not** a gland. The module-end USB-B plug is retained by a printed P-clip screwed to a POD-P1 boss 40 mm behind the connector so the plug cannot back out inside a closed enclosure. At service the whole pigtail is a replaceable item: undo the P-clip, undo the two receptacle screws, fit a new one. That is the replaceable element the DSN-

EEG-002 section 9 risk 9 mitigation depends on, and it replaces the gland insert of Rev A.

The participant's cable is not part of this document. RFQ A-07 ships two 1.0 m cables with every kit, one USB-C to USB-A and one USB-C to USB-C, plus a 5 V 2 A EU charger. **One of those two is the host lead** and plugs into the panel receptacle above; the other serves the charge port at WH-07. Which of the two is which depends on the participant's computer, which is exactly why two are shipped. Rev A of this document said both A-07 cables served the charge port and that the host link was captive; that is corrected.

The DevKit's own USB-C receptacles. The two USB-C receptacles on the ESP32-S3-DevKitC-1 sit on the patient-applied side of the barrier and are brought to no panel aperture. They are reachable only with the lid off, through the 31 x 61 mm opening in MP-01 over the DevKit, which is where end-of-line flashing is done: the DevKit's UART USB-C port carries the auto-reset circuit and the carrier's J26 header is console and recovery only, because GPIO0 is committed to LED_SR_LATCH. This is an assembly requirement in ASM-EEG-007, repeated here because the harness builder is the last person who sees those receptacles before the plate goes on.

3.9 The five panel adapters

Rev A named WH-ADP-01, -01B, -02, -03 and -04 in five wire lists and specified none of them, so five of the eight assemblies terminated on a part number with nothing behind it. Each is settled here: what it mates at each end, what it is made of, its own wire list, and what has to be confirmed before it can be bought or printed. Two are bought parts and three are printed, and the printed three are modelled in `tools/mech_gen.py` and released as STEP and STL with the rest of the print set. **None of the five has been made or measured.**

Adapter	Cable	POD-P1 opening, design (mm)	What it is
WH-ADP-01	WH-03, boom microphone	6.5 dia at 122.0, 90.0	a bought jack; no printed part
WH-ADP-01B	WH-04, headphones	6.5 dia at 128.0, 72.0	the same bought jack
WH-ADP-02	WH-05, room microphone	4.0 dia at 122.0, 102.0	a printed carrier and an unqualified module
WH-ADP-03	WH-07, charge port	10.0 x 4.0 at 143.0, 80.0	a printed clamp plate and a bought receptacle
WH-ADP-04	WH-09, host port	10.0 x 4.0 at 146.0, 12.0	a printed clamp plate and a bought receptacle

Those are the openings `tools/mech_gen.py` cuts, and they are all on the **right-hand wall** at 45 % of the enclosure height. **Four of the five overlap a button opening.** The three response buttons are 13.0 mm openings on the same wall at the same height, at design y = 76, 90 and 104 mm, so BTN_B is concentric with the boom opening, BTN_STOP contains the room-microphone port, BTN_A overlaps the headphone opening, and the charge opening runs into both BTN_A and BTN_B. That is a fault in `pod_base()`, not in these adapters – they are dimensioned to the openings as they are specified – and no adapter here can be fitted until it is corrected. It is open item 20.

WH-ADP-01 and WH-ADP-01B – the 3.5 mm panel jacks. Bought.

What it mates. Outside: the moulded 3.5 mm 4-conductor plug of WH-03B on WH-ADP-01, the participant's headphone plug on WH-ADP-01B. Inside: the four flying leads of WH-03 or WH-04, soldered to the jack's own lugs. Nothing else.

What it is made from. A 4-conductor 3.5 mm panel-mount jack with a threaded barrel, M6 x 0.75, for a 6.5 mm panel hole and a panel up to 2.5 mm thick, with solder lugs – the CUI SJ-435xx and Lumberg KLBR classes are both of that form. It is fitted with its own nut and a 10 mm outside diameter, 0.8 mm silicone washer behind the flange. **There is no printed part**: POD-P1's wall is 2.5 mm and the jack's own thread and nut hold it, so a bracket would add a part without adding a function.

Wire list. Sections 3.3 and 3.4, one conductor to each of tip, ring 1, ring 2 and sleeve, with the WH-03 drain cut back 10 mm at the jack and insulated. It is not repeated here.

What has to be confirmed, and none of it is. Which lug is which contact on the part actually bought, because the tip / ring 1 / ring 2 / sleeve lug order is not common across the class and getting it wrong puts DGND on VOICE_RAW; that the barrel is at least 4.5 mm long, so that it passes a 2.5 mm wall and still takes its nut; that the part tolerates POD-P1's **printed** 6.5 mm hole, which an MJF part holds to about +/-0.3 mm and which is not a machined panel hole; and the plug retention, against the 30 N boom-detach limit of test H6. On WH-ADP-01B there is one more: section 3.4 uses the fourth contact as the insertion detect that J27.4 reserves, and **a switched detect contact is not a feature every jack in this class carries**. If the part bought has none, J27.4 stays a no-connect, the detect of section 3.4 is not implemented, and this document does not pretend otherwise.

WH-ADP-02 – the room-microphone carrier. Printed, and it carries a module that does not exist.

What it mates. Outside: bonded to the inside of the POD-P1 wall over the 4.0 mm acoustic port, with a silicone washer in the 10 mm recess on its wall face. Inside: the capsule and switch module, on M2.5 standoffs, sealed to the plate by a second washer in the recess on the module face. Cable: the four conductors of WH-05 and its drain, at the module.

What it is made from. `tools/mech_gen.py wh_adp02()`, MJF PA12, **32.0 x 24.0 x 3.4 mm, 1.96 cm3** *(model figures, read from the released mesh; the 3.0 mm of the first issue was the plate thickness without the gasket-recess boss)*: a 4.0 mm acoustic port, a 10 mm gasket recess on each face, six M2.5 clearance holes on an 8 mm grid and two 6.0 x 2.5 mm tie slots. The grid is deliberate, and it is the answer MP-01 gives for the same reason: **no room-microphone module is qualified** (section 3.5 and open item 5), so no hole pattern is known, and a plate drilled to a guessed pattern is worse than a grid that a small breakout picks up two holes of and a tie holds the rest of the way.

Wire list. Section 3.5 gives the four conductors at J28. On the adapter itself:

Node	From	To
supply	WH-05 conductor 1, DVDD3V3	capsule V+ and the analogue switch V+
return	WH-05 conductor 2, DGND	capsule ground, switch ground, and the WH-05 drain
audio	capsule output	the switch’s signal input
audio out	switch common	WH-05 conductor 3, ROOM_PRE
mute	WH-05 conductor 4, MIC_MUTE	the switch’s control pin, with a 100 kOhm resistor holding it in the **muted** state

That 100 kOhm sits on the adapter and not on the carrier, and it is there so the room microphone is muted whenever GPIO21 is an input – which it is at reset, before any firmware runs, which is when a microphone in a participant’s home is least supervised. **Which logic level mutes is not fixed by this document**: it follows from which throw of the switch the sub-assembly uses, and the sub-assembly has no drawing.

What has to be confirmed. Everything about the module. RFQ E-15 asks for a mute that is a hardware gate in the signal path and no catalogue part is known to meet it. The candidate is a Knowles SPU0414HR5H-SB class analogue MEMS capsule with a TI TS5A3159 analogue switch in the output path; the fallback is that same pair as a programme-designed sub-assembly, which needs a drawing, a part number and an AVL line and has none of the three. This carrier is dimensioned so that either of them lands on the same plate, which is the most this document can do while the part is open.

WH-ADP-03 and WH-ADP-04 – the USB-C panel plates. Printed, plus a bought receptacle.

What they mate. Outside: one of the two A-07 cables the kit ships. Inside: WH-07’s two conductors on WH-ADP-03, WH-09’s four conductors and its braid on WH-ADP-04.

What they are made from. `tools/mech_gen.py wh_adp_usb()`, MJF PA12. WH-ADP-03 is **34.0 x 20.0 x 5.0 mm, 2.18 cm3**; WH-ADP-04 is **34.0 x 20.0 x 7.4 mm, 2.65 cm3** *(model figures)*. A USB-C receptacle nose fills the 10.0 x 4.0 mm opening, so no part of the plate can sit in it and the plate works from behind: a picture-frame rim lands on the inside of the wall, the receptacle’s mounting flange is trapped inside that rim, and a 14.0 x 8.0 mm window passes the receptacle body and its wires. The rim takes **any** flange up to 24.0 x 14.0 x 1.6 mm rather than being cut to one receptacle, because no receptacle has been bought and a pocket cut to a guessed flange is a plate that fits nothing.

WH-ADP-04 differs in two ways and both of them are the isolation barrier: its rim is 0.8 mm deeper, for the flange gasket section 3.8 calls for, and a 2.0 mm skirt around the window adds 4.0 mm to the surface path from the receptacle’s own terminals to anything bonded to the pod wall. PA12 is not conductive and neither plate carries a metal insert, so nothing in either of them can become the bond the ADuM4160 exists to prevent.

The receptacle. A panel-mount USB-C 2.0 receptacle with a two-hole flange and solder cups or a short pigtail, carrying **two 5.1 kOhm CC pull-downs** so that a Type-C source or host sees a device. The pull-downs are fitted on the adapter, at the receptacle's own CC cups, under heat-shrink; they are not on the carrier and no carrier net reaches them.

Wire list, WH-ADP-03, the charge port. WH-07 is section 3.7.

Receptacle	To	Note
A4, A9, B4, B9 commoned	WH-07 conductor 1, VBUS_IN, to J24.1	
A1, A12, B1, B12 commoned	WH-07 conductor 2, DGND, to J24.2	
CC1 (A5)	5.1 kOhm to the commoned ground	on the adapter
CC2 (B5)	5.1 kOhm to the commoned ground	on the adapter
D+, D-, SBU1, SBU2	no connection	**no data conductor enters WH-07**
shell	no connection	isolated from DGND; that ruling is open item 8

Wire list, WH-ADP-04, the host port. WH-09 is section 3.8.

Receptacle	To	Note
A4, A9, B4, B9 commoned	WH-09 VBUS, to the module's USB-B pin 1	host side only; no path to the battery rail
A6, B6	WH-09 D+, to USB-B pin 3	
A7, B7	WH-09 D-, to USB-B pin 2	
A1, A12, B1, B12 commoned	WH-09 GND, to USB-B pin 4	host-side 0 V, **not DGND**
CC1 (A5), CC2 (B5)	5.1 kOhm each, to that host-side 0 V	**not to DGND**
shell	WH-09 braid, to the USB-B shell	bonded at both ends of WH-09 and to nothing else

Every node in the WH-ADP-04 table is on the host side of the barrier, the two pull-downs and the shell included, and test H10 is what measures that, at 500 V DC, with WH-09 unplugged.

What has to be confirmed. That the receptacle's nose stands at least 2.5 mm proud of its flange – 3.3 mm on WH-ADP-04, which has a gasket in front of the flange – so that it reaches through the wall; that the flange is inside 24.0 x 14.0 x 1.6 mm; and that its CC cups are reachable for the pull-downs once it is in the rim.

And how the plates are held on. All three are bonded to the inside of the wall, because `pod_base()` carries no boss for WH-ADP-02, WH-ADP-03 or WH-ADP-04. Both USB plates carry two M2.5 clearance holes for the screws that would replace the bond, and there is nothing in POD-P1 for those screws to go into. Test H6 pulls the panel receptacle at 50 N: that is then a pull on a bonded joint which has not been made and has not been tested. The section 3.8 P-clip has the same gap – it is specified as screwed to “a POD-P1 boss 40 mm behind the connector” and WH-09 has no such boss. Open item 21.

---

4. Materials

Property	WH-01	WH-02	WH-03, WH-04, WH-05	WH-03B	WH-07
Conductors	11 signal + drain	10	4	**2 signal + drain**	2
Conductor	7/0.1 mm tinned copper	7/0.1 mm tinned copper	7/0.1 mm tinned copper	**7/0.1 mm tinned copper**	7/0.2 mm tinned copper
Nominal CSA (calculated)	0.055 mm2	0.055 mm2	0.055 mm2	**0.055 mm2**	0.22 mm2
DC resistance (calculated, 0.0172 Ohm.mm2/m)	0.313 Ohm/m	0.313 Ohm/m	0.313 Ohm/m	**0.313 Ohm/m**	0.079 Ohm/m
Primary insulation	PTFE, 0.15 mm wall, OD 0.70 mm	PVC, 0.18 mm wall, OD 0.75 mm	PTFE, OD 0.70 mm	**PTFE, 0.15 mm wall, OD 0.70 mm**	PVC, OD 1.15 mm
Colours	section 3.1	section 3.2	sections 3.3 to 3.5	**RD capsule +, BK capsule -**	section 3.7
Screen	overall Al/PET foil, 100 % coverage, + 30 AWG tinned drain	none	overall Al/PET foil + 30 AWG drain	**overall Al/PET foil, 100 % coverage, + 30 AWG tinned drain**	none
Jacket	TPU, 0.55 mm wall, OD 4.3 mm nom / 4.6 mm max	TPU, 0.50 mm wall, OD 4.5 mm nom	TPU, OD 2.2 mm max	**TPU, OD 2.2 mm max**	PVC, OD 3.0 mm

Property	WH-01	WH-02	WH-03, WH-04, WH-05	WH-03B	WH-07
Min bend radius	26 mm dynamic, 13 mm static (6x / 3x OD)	27 mm / 14 mm	14 mm / 7 mm	**14 mm / 7 mm**	18 mm / 9 mm
Single conductor min bend radius	5 mm	5 mm	5 mm	**5 mm**	8 mm
Temperature	-20 to +200 C (PTFE)	-20 to +80 C	-20 to +200 C	**-20 to +200 C (PTFE)**	-20 to +80 C
Chemical	resists 70 % IPA and conductive EEG paste	as WH-01	as WH-01	**as WH-01, and repeated 40 kHz ultrasonic immersion**	as WH-01
Flex life target	50 000 cycles at the crown transition	50 000 cycles	not flexed in use	**a duty, not a cycle count – see below**	not flexed in use
Cut length	section 3.1	section 3.2	220 / 190 / 180 mm	**1700 mm, +10 / -0**	150 mm

WH-09 is a bought-in shielded USB 2.0 assembly and is specified by its USB conformance, not by the table above. **WH-10 is a bought-in lead of the same class as the three WH-06 EMG leads and is specified in section 3.1.3, not here.**

Insulation of the three free tails, new on 2026-09-02. Conductors 9, 10 and 11 leave the jacket and the screen at the helmet end and run free – 250 mm at each temple and F mm at the halo front. Over that whole free length each carries a **secondary sleeve over its PTFE primary**: extruded PTFE or FEP, or 3:1 heat-shrink supplied at 1.5 mm or more and **recovered to 0.70 mm or less** so that it grips the conductor rather than tenting over it, resistant to 70 % IPA and to the 40 kHz ultrasonic bath the clips go into. It is a materials requirement and it is written here for that reason. *A printed marker sleeve is not insulation and must not be cited as this part: the Brady PS-187-2-WT class of section 10 recovers to 1.6 mm, more than twice the 0.70 mm OD of the conductor it would have to insulate.* AVL-EEG-017 K34 buys it.

WH-03B, and why it now has a column. Rev B registered the boom lead in section 1, gave its four connections in section 3.3, and left it out of this table. The one cable in the kit that a participant bends by hand at the start of every session therefore had no conductor construction, no gauge, no insulation, no colour code, no screen, no jacket, no OD, no bend radius and no temperature or chemical rating – and a harness shop could neither cut it, buy it, nor lay it up. It is specified above as the two-conductor member of the WH-03 / WH-04 / WH-05 family rather than as a new construction, because electrically it **is** the continuation of WH-03: the same 7/0.1 mm conductor, the same PTFE primary, the same foil-and-drain screen and the same TPU jacket, so it adds no new raw material to the kit and no new qualification. Three things follow from where it goes, and they are stated rather than assumed.

- **Its screen is terminated at both ends**, which no other screened cable in this document does. Section 3.3 lands the screen on ring 2 and the sleeve of the 3.5 mm plug, and the capsule end bonds it to the capsule can. That is deliberate and it is not a breach of the single-point rule of section 5.1: both ends of this cable are at DGND through the jack anyway, the lead is outside the frame and outside the pod for its whole length, and it is **not part of the helmet screen system** and never reaches R91.
- **It is immersed.** SVC-EEG-013 section 3 R5 puts the boom in the 40 kHz ultrasonic bath at every turnaround. The PTFE primary and the TPU jacket are chosen for that; **a PVC jacket is not acceptable on this assembly and no deviation may substitute one.**
- **Its plug is assembled, not moulded.** Section 3.3 called for “a moulded 3.5 mm 4-conductor plug”. An over-moulded plug cannot be fitted in a harness shop and cannot be replaced at service, and this is a lead that test H6 pulls to 30 N at the participant-facing detach point. The word is corrected here: the plug is a **4-pole 3.5 mm plug with solder buckets and a screwed or crimped strain-relief barrel**, so the assembly can be built, pull tested, and repaired. AVL-EEG-017 K41 carries it.

Its flex duty is a duty, not a cycle count. WH-01 and WH-02 carry a 50 000-cycle target at the crown transition. No equivalent number is stated for WH-03B, because nothing in the package says how many times a boom is set in a unit’s life, and a cycle count invented to fill the cell would be worse than the empty cell it replaced. What is stated instead is the duty: the lead is worked at the gooseneck root every time the boom is positioned, and it must survive the gooseneck’s full articulation with no continuity change. **Test H7 is run at the gooseneck root rather than at the crown for the WH-03B first article.** The number belongs to that first article – open item 25.

On the gauge designation. 7/0.1 mm has a calculated cross-section of 0.055 mm2, which is 30 AWG by area and is catalogued as “28 AWG equivalent” by several European cable houses because its overall diameter matches 28 AWG. Where a supplier’s 28 AWG is 7/0.127 mm (0.0887 mm2, 0.194 Ohm/m) that construction is the preferred alternate and needs no deviation: it is lower resistance and the same OD class. No other substitution is permitted without a written deviation, because the OD drives the channel fill in section 7.

Why the resistance does not matter, and where it does. The longest WH-01 run is REF_L at 2140 mm: 0.67 Ohm calculated, against a 47 kOhm series resistor and a 5 to 20 kOhm contact impedance. It contributes 14 parts per million of the source impedance and is not a signal consideration; the 1.0 Ohm continuity limit in section 9 exists to catch a bad crimp, not to protect the measurement. On WH-02 the LED current of 1.3 mA drops 0.9 mV in the longest run and cannot shift a colour boundary.

Why WH-01 is screened and WH-02 is not. The eleven electrode conductors are high-impedance nodes referenced to AGND_REF, the analogue mid-rail, carrying microvolts in a 0.5 to 70 Hz band against the input-referred noise budget of RFQ E-03. The arithmetic of that budget – the Johnson noise of the series resistors, the converter contribution and the flatness at 100 Hz – is worked once, in RISK-EEG-011 section 4, and is not recomputed here; what it leaves is very little room for pickup. These conductors run beside a person for two hours in a domestic room full of mains field. They are screened. The eight light conductors are driven from a 74HC595 through 1 kOhm into a 2 V LED: source impedance under 1.1 kOhm, no high-impedance node anywhere, and the lights are forced dark for the whole of every recording block.

Corrected 2026-09-02: that last clause used to read “so the cable carries no switching current at any moment when an electrode channel is live”, and it is not true. RFQ-EEG-001 E-27 drives the contact lights *from the converter’s own lead-off measurement*, so a lit light means a live measurement on the same electrode by definition. What the recording interlock gives is narrower and still worth having: no switching current during a **recording block**. The concurrent case is electrode preparation – the participant gelling a site and watching its light change – and that is not an edge case, it is the state the lights exist for. It is also the case in which the LED and the electrode conductor are closest together, because at the site they share the HM-04 body and the frame’s 6 mm channel separation does not apply there; the 47 kOhm series resistors are on the carrier, so the electrode conductor runs at full electrode impedance the whole way from the cup to J14.

The controls below are real and are unchanged. What was missing was a limit and a measurement: nothing in the package stated how much noise the lights were allowed to add, and no test measured it – TST-EEG-004 T9a measures channel-to-channel crosstalk on the carrier through a fixture, with no harness and no lights. **RFQ-EEG-001 E-30 now states the limit and TST-EEG-004 T9c measures it, through the fitted harness in an assembled helmet.** Until T9c has been run on a built unit, the decision to leave WH-02 unscreened rests on the argument in this section and not on evidence. A screen would earn nothing and would cost a second drain, for which there is exactly one shield pin on the whole carrier. The separation that matters for WH-02 was made physically by ECO-EEG-014, which moved it to J30 in the digital zone across the x = 62 mm split, and by the two-channel routing of section 7.

---

5. Screen and drain

5.1 The single-point rule

The star-point rule itself is stated once, in DSN-EEG-003 section 3.3, and this document cites it rather than restating it. Its consequences for the harness are these, and they are what the builder is signing for:

`HARN_SHIELD` reaches DGND at **R91 only**, a 0 Ohm 0603 at design coordinate (56.0, 62.0), fab (56.0, 68.0). AGND_REF reaches DGND at **R90 only**, at design (56.0, 58.0), fab (56.0, 72.0). Fit exactly one of each. Never bridge either with a wire link or a solder blob, and never fit a second drain anywhere in the kit that reaches DGND by another route. Two star points in parallel is a ground loop around the whole helmet, and its signature – 50 Hz pickup that appears on every channel at once – will be blamed on the electrodes. On the four-layer carrier there are now two reference planes on each side of the split rather than one, and R90 and R91 remain the only ties between the AGND_REF planes and the DGND planes; the stitching vias tie L2 to L3 within one reference, never across the split.

5.2 WH-01 construction

- **Screen:** aluminium/polyester foil, 100 % coverage, applied over the eleven conductors with a 25 % overlap, foil side inward against the drain.
- **Drain:** one 30 AWG tinned-copper conductor laid in continuous contact with the foil, in the interstice, running the full length.

- **Pod end:** foil and drain terminated together. The drain is sleeved in clear 2:1 heat-shrink from the jacket cut to within 6 mm of the crimp, crimped into a single contact, and inserted in position 12 of the WH-01 housing. One drain, one pin. There is no splice.
- **Helmet end:** foil and drain cut back 10 mm inside the jacket, folded back on themselves and covered with 15 mm of 2:1 adhesive-lined heat-shrink. **The screen is not connected to the cup, to the spring, to the HM-04 body, to any LED, to WH-BUS-01 or to the frame.** This is a “do not connect” instruction, and section 9 test H3 exists to prove it was followed.
- Inside the frame each conductor continues unscreened. That is a deliberate ruling: the frame is a closed PA12 monocoque with no conductive part, the in-frame runs are 130 to 315 mm, and an individually screened run cannot be drawn through a 3.8 mm channel and replaced through a cover strip, which is the service story DSN-EEG-002 section 4.1 depends on. The residual risk is mains pickup on the unscreened section and it is verified, not assumed, at the input-referred noise step of TST-EEG-004 (inputs shorted, 60 s at 1 kHz, 1.0 uV RMS maximum in 0.5 to 70 Hz).

5.3 WH-03, WH-04, WH-05

Each carries its own overall foil and 30 AWG drain, terminated at the **carrier end only**: the WH-03 and WH-05 drains on the **DGND pin** of their connectors (J18.2, J28.2), and the WH-04 drain on **J27.3 = HP_GND**, the headphone return, which is referenced to DGND inside the codec module and, since the routing closed on 2 September 2026, **reaches DGND through the carrier copper**; `kicad/EEG-CAR-01_RevB_DRC_report.txt` records all 145 nets fully connected with none unclosed and none without copper, HP_GND (J8.13 to J27.3) among them, so the WH-04 drain no longer floats and this harness needs no change. At the far end the drain is cut back 10 mm and insulated. The WH-03 and WH-05 drains reach DGND directly and do not pass through R91; all three are entirely inside the pod, in the digital zone, and are not part of the helmet screen system. The WH-09 braid is not part of either system and is bonded only within WH-09 itself, per section 3.8.

5.4 WH-02

No screen, no drain. LED_GND is the guard conductor and is described in section 3.2. The justification for carrying no screen is in section 4, together with the correction of 2026-09-02 to the sentence that used to justify it, the E-30 limit that now bounds it and the T9c measurement that will decide it.

5.5 Leakage

RFQ S-02 allows a maximum of 10 uA DC and 100 uA AC through any electrode in normal condition and 50 uA DC in single fault. In normal condition the screen path contributes nothing: it is tied to DGND, the electrode conductors are separated from it by PTFE tested to at least 100 MOhm at 500 V DC, and the working voltage across that insulation never exceeds the +/-2.5 V analogue rails. The calculated worst-case leakage through the insulation at 2.5 V across 100 MOhm is 25 nA per conductor, three orders of magnitude below the normal-condition limit.

The single-fault limit of S-02 is not met, and this document does not claim it is. With one conductor shorted to the drain the 47 kOhm series resistor is still in circuit and the fault current is bounded by the rails at 2.5 V / 47 kOhm = **53.2 uA against the 50 uA limit of S-02**. The fix is ECO-EEG-024, which raises R1 to R16 from 47 kOhm to 68 kOhm and gives 2.5 V / 68 kOhm = 36.8 uA; that keeps the input corner at 234 Hz, which is -0.75 dB at 100 Hz and therefore requires RFQ E-10 to widen to +/-1.0 dB, and raises the resistor noise to 0.28 uV for a total of 0.31 uV, still well inside E-03. **The Phase 1 prototypes are built with 47 kOhm fitted**, the measurement is made on them, and the change is made before Phase 2. Until then S-02 single fault is a stated non-conformance carried for the safety reviewer, not a compliance. The primary means of compliance remain R1 to R16 and the ADuM4160 isolator, per E-07 and E-24.

---

6. Connectors, contacts and tooling

J14 and J30 are 2.54 mm **socket** strips on the carrier (Samtec SSW-112-01-G-S and SSW-110-01-G-S or equivalent), 1.00 mm finished plated holes. The harness therefore terminates in male crimp contacts. Pin 1 of every socket strip is the square pad and is marked on the top silkscreen.

Item	Board side	Harness side	Contact	Tool	Notes
WH-01 to J14	1x12 socket strip, 2.54 mm	Harwin M20 12-way male crimp housing, polarised	Harwin M20-118 male crimp, gold flash, 24-30 AWG	Harwin Z20-320 hand tool; Z20-431 extraction	confirm the current Harwin ordering suffix at IQC and record it in AVL-EEG-017
WH-02 to J30	1x10 socket strip	Harwin M20 10-way male crimp housing	as above	as above	
WH-03 to J18	1x4 socket strip	Harwin M20 4-way male crimp housing	as above	as above	
WH-04 to J27	1x4 socket strip	as above	as above	as above	
WH-05 to J28	1x4 socket strip	as above	as above	as above	
WH-06 to J15-J17	touch-proof 1.5 mm socket, **no confirmed part** (section 3.6)	Staubli SLS425-SEK/N touch-proof plug	moulded, bought-in	none	1.70 mm plated hole, 1.50 mm NPTH retention posts
WH-07 to J24	JST B2B-PH-K-S(LF)(SN)	JST PHR-2 housing	JST SPH-002T-P0.5S	JST WC-160 / YC-160R	2.00 mm pitch, 0.90 mm plated holes
WH-09	none – module USB-B receptacle to panel	moulded USB-B plug; panel USB-C receptacle on WH-ADP-04	–	–	plug retained by printed P-clip; receptacle gasketed to the panel

The **helmet-end and boom-end** terminations, which Rev B named and did not specify:

Item	Fixed side	Removable side	Contact	Tool	Notes
WH-01 conductors 1-8, at the eight sites	**HM-04A** termination contact, anchored in the HM-04 body, conductor soldered to its tail 8 mm outside the body	**HM-05C** contact crown on the HM-05B spigot; parts with the bayonet	axial, gold on gold, 0.5 N minimum	soldering iron; HM-09 service key	**PROPOSAL, section 3.1.1.** No part, no vendor and no geometry today. AVL-EEG-017 K25 and K26
WH-01 conductors 9 and 10, at the ears	free-hanging 1.5 mm touch-proof socket to **DIN 42802-1**, GY and PK bodies, on the 250 mm temple tail	the K2 ear clip's own DIN 42802 touch-proof plug, colour matched or sleeved within 25 mm of the plug	crimp or solder bucket for 28 AWG; **separation force 5 to 15 N**, mating force stated, 500 cycles minimum	per the coupler maker	**RULED 2026-09-02, section 3.1.2.1; not signed.** AVL-EEG-017 K27, ordered on the same purchase order as K2
WH-01 conductor 11, at Fpz	free-hanging 1.5 mm touch-proof socket to **DIN 42802-1**, TQ body, at the HM-01 halo-front mouth on a stated free tail F	**WH-10**: 150 mm +/- 10 mm snap-to-DIN-plug lead, TQ, onto a K4 disposable pad	crimp or solder bucket for 28 AWG; retention **13 N maximum** until HM-01 gains an anchorage, not H6's 15 N	per the coupler maker	**RULED 2026-09-02, section 3.1.3; not signed.** AVL-EEG-017 K27 third unit and K47. *Was: "bias pad, solder tag" – no part and no drawn feature*
WH-02, both leads at each of the eight site LEDs	two-lead bicolour LED, ASM-EEG-007 section 4.3	–	soldered and sleeved	soldering iron	the LED has no seat and no lead passage in HM-04 – section 3.1.1

Item	Fixed side	Removable side	Contact	Tool	Notes
WH-03B, at the capsule and at the panel	electret capsule solder tags	4-pole 3.5 mm plug, solder buckets, screwed or crimped barrel	–	soldering iron	**not over-moulded** – section 4. AVL-EEG-017 K41

One connector system, and which one it is. Every crimped 2.54 mm connector in this kit – the five harness housings above and the seventeen module jumpers of ICD-EEG-006 section 3 – is the **Harwin M20 crimp system**. That has to be said in one place because until this issue the package said it in three, incompatibly: this section's M20 rows, AVL-EEG-017 section 1.6's "Harwin M20-106 series or 3M 89 series", and ICD-EEG-006 section 3.1's "a Molex KK 254 crimp build: female housings 22-01-30nn with 08-50-0114 terminals at the module end, free-hanging male housings with 08-52-0072 terminals at the carrier end". The choice is settled by geometry and not by preference: `tools/mech_gen.py` cuts the WH-KEY-01 cavity at `KEY_CAV_W = M20_HSG_W + 0.30 = 4.50 mm` and asserts both that the housing enters with 0.30 mm of clearance and that a reversed housing is stopped by `M20_HSG_W + M20_KEY_MIN - KEY_CAV_W = 0.40 mm` of interference. **A housing outside 4.20 +0.10 / -0.20 mm across the flats either will not enter the shroud or will not be keyed by it**, so adopting a different family would have made every printed shroud in the kit the wrong size, silently, and the only mitigation against the one safety-relevant mis-mate in the kit would have been a dust cover. **ICD-EEG-006 section 3.1 is corrected to the M20 system at this issue, and AVL-EEG-017 section 1.6.1 now carries the per-way-count housing and contact table that both documents used to point at and neither contained.** The 4.20 mm itself is still UNCONFIRMED and is measured at IQC before a shroud is printed – open item 11.

Polarisation and mis-mating. ECO-EEG-014 removed some of the risk by construction: J14 is 12-way and J30 is 10-way, so WH-01's housing cannot enter J30. It removed less than Rev A claimed, because a 10-way housing does fit a 12-way socket and WH-02 can therefore reach J14. What is left is that, offset and reversal on a bare socket strip, and the mating housings alone prevent none of it. The anti-reversal feature is the printed shroud **WH-KEY-01**, part of the MP-01 print set, which surrounds J14 and J30 and takes the housing's polarising rib in one orientation only; it is dimensioned in section 6.1, which also states the one of these four mis-mates it does not stop. A third form of WH-KEY-01 exists for J22; it is fitted only when the Phase 2 EOG panel option is taken, since no cable in this register lands on J22 (section 2). This is the same keying decision ICD-EEG-006 section 6 records for the module jumpers: a shrouded polarised header at the module end where the module has one, and the printed carrier-end shroud over every socket that takes a jumper. The shroud names SHR-14-A, SHR-30-A and SHR-22-A of Rev A of JIG-EEG-009 are withdrawn: JIG-EEG-009 Rev C section 1.10 calls all three WH-KEY-01, and PARTS-EEG-019 records the old names as legacy. The consequences if the shroud is omitted:

Mis-mate	Result
WH-01 offset +1	E_Fz onto the Cz channel, every site reported one place out, HARN_SHIELD floating and the screen open
WH-01 offset -1	BIAS_EL onto REF_R; the driven common-mode return lands on the reference node
WH-01 reversed	E_Fz onto HARN_SHIELD, BIAS_EL onto E_Cz; a driven output onto a protected scalp input
WH-02 reversed	LED_GND onto LED1 and LED_V onto LED2; lights wrong, no safety consequence

Only WH-01 has a safety-relevant mis-mate, and in every case the 47 kOhm series resistors and the +/-2.5 V rails still bound the current – to the 53.2 uA of section 5.5, which is over the single-fault limit. That is a mitigation with a stated shortfall, not a licence: WH-KEY-01 is mandatory.

Insertion force. Supplier data for 2.54 mm gold-flash contacts is 0.6 N typical and 1.5 N maximum per contact. Calculated totals: 12-way 7.2 N typical, 18 N maximum; 10-way 6.0 N / 15 N; 4-way 2.4 N / 6.0 N. Withdrawal force after 25 mating cycles must remain at or above 0.15 N per contact. Nothing here has been measured.

Strain relief and service loop. Each helmet cable is clipped to a POD-P1 boss with a printed P-clip 40 mm behind its housing, so that any pull is taken by the enclosure and not by the crimps. Behind the clip, **120 mm of service loop** is coiled at 30 mm diameter and tied with a hook-and-loop strap, not a cable tie – it must be undone by hand at service. The loop exists so that a replacement conductor can be drawn through a channel and terminated without dismantling the far end, which is the requirement in DSN-EEG-002 section 6 and the mitigation in section 4.1. A further **40 mm service loop** is coiled at each HM-04 assembly for the same reason at the other end. **Conductor 11 carries no site service loop from 2026-09-02**: its site is a coupler on a free tail and there is nothing to coil it at, which is why its cut length is 1940 + F and not 1980 mm (section 3.1.3). Conductors 9 and 10 keep theirs – the 40 mm is inside their 2140 mm and is coiled at the temple before the tail runs free. The enlarged POD-P1 internal volume of 158.0 x 138.0 x 55.5 mm is what these loops are coiled into, and the carrier now stands on M3 x 18 mm nylon standoffs above the floor, so the loops sit under the plate and not against the carrier edge.

6.1 WH-KEY-01, dimensioned

Rev A named the shroud, made it mandatory, and drew nothing. It is now modelled, in `tools/mech_gen.py wh_key01()`, in the three forms JIG-EEG-009 Rev C section 1.10 asks for, and it is released as STEP and STL with the rest of the MP-01 print set. MJF PA12. Coordinates are the design convention of section 1. **Every figure here is from the model; nothing has been printed and no housing has been in a shroud.**

Form	Socket	Ways	Outside (mm)	Volume	Keyway side	Footprint on the carrier
WH-KEY-01/ J14	J14	12	8.30 x 33.68 x 12.50	1.57 cm3	+X	x 1.15 to 9.45, y 9.33 to 42.61
WH-KEY-01/ J30	J30	10	8.30 x 28.60 x 12.50	1.36 cm3	-X	x 61.55 to 69.85, y 87.33 to 115.53

Form	Socket	Ways	Outside (mm)	Volume	Keyway side	Footprint on the carrier
WH-KEY-01/ J22	J22	3	8.30 x 10.82 x 12.50	0.57 cm3	+X, which is board +Y at J22	x 22.25 to 32.67, y 112.15 to 120.45

It is a rectangular tube, open top and bottom, that drops over a socket strip already soldered to the carrier and stands 4.00 mm proud of it. Walls are 1.20 mm, and 2.20 mm on the wall that carries the keyway. Everything that could be derived was: the cavity is 4.50 mm wide by N x 2.54 + 0.40 mm long because `fplib.pinsocket_1xn` makes the socket body 2.54 mm wide on a 2.54 mm pitch, and the 12.50 mm overall height is the **8.50 mm of socket mating height that ICD-EEG-006 section 4 budgets** plus 4.00 mm of lead-in, so that the shroud has hold of the housing before the male contacts reach the socket. One pair of internal ribs on the J22 form and two on the others close to 2.50 mm on the socket’s 2.54 mm body, below the mating face where they can never touch the housing, and locate the shroud; a bead of adhesive at the rim holds it. MJF holds about +/-0.30 mm on a feature this size, so the ribs are a location and the adhesive is the retention, and neither has been tested.

Which side the keyway is on is a board-clearance decision, not a preference, and it is recorded per form because the housing has to match it. At J14 the fiducial FID1 sits at x = 10.0 mm and the board edge at x = 0, so the keyway goes on +X. At J30, R70’s pad starts at x = 70.53 mm, so it goes on -X. Calculated clearances, from `design.py` and the model, to the KiCad footprint envelopes in `tools/fplib.py`, which include the courtyard: the J14 form is 0.55 mm from FID1’s and 1.33 mm from MH1’s; the J30 form is 0.68 mm from R70’s, which is 0.91 mm to R70’s copper, and reaches y = 87.33, which is **0.44 mm from J9’s socket body**. J9 is one of the seventeen module sockets ICD-EEG-006 section 6.1 also puts a WH-KEY-01 on, and 0.44 mm will not hold a second end wall, so either the 1x4 module form is open-ended on that side or J9 moves. The module forms are not modelled here and that is open item 19.

What it blocks, and what it does not.

Mis-mate	Blocked	By what
WH-01 reversed end for end	yes	the keyway is a 2.60 x 1.00 mm slot in one long wall, over way 1 only. The right way round, the housing’s polarising rib slides down it; turned end for end the rib meets a flat wall and the housing then needs 4.90 mm of a 4.50 mm cavity
WH-01 or WH-02 offset by one way	yes	the cavity is the housing length plus 0.40 mm and is closed at both ends. A pitch is 2.54 mm
WH-01’s 12-way housing into J30	yes	30.5 mm of housing into a 25.8 mm cavity

Mis-mate	Blocked	By what
WH-02’s 10-way housing into J14	**no**	25.4 mm of housing enters a 30.9 mm cavity, either way round. Way count does not stop it either: a 10-way housing fits a 12-way socket, so this section’s claim that the two helmet cables “physically cannot be swapped” holds in one direction only. What catches this one is the pin-by-pin read-back at step 8 and test H9, not this part

The two dimensions this part does not have. The shroud is cut for a Harwin M20 male crimp housing **4.20 mm wide across the flats** whose **male contacts stand 4.00 mm proud of the mating face**, and neither figure is in this package and neither has been measured. They are `M20_HSG_W` and `M20_PIN_PROUD` in `mech_gen.py` and the part regenerates from them. A cavity 0.30 mm narrower than the housing will not take the cable at all, and one 0.50 mm wider stops keying it, so **the housing is measured and the shroud reprinted before a build**, and the IQC step this section already requires for the Harwin ordering suffix records the housing envelope with it.

And one dimension it imposes. The shroud keys nothing unless the housing carries a **polarising rib at least 0.70 mm proud of its body, within the first way from the way-1 end**. Section 6 specifies a polarised housing; where that housing’s rib actually is has not been confirmed. If the part bought has no rib, or carries it elsewhere, WH-KEY-01 is a dust cover, the reversed WH-01 of the table above is stopped only by the read-back at step 8 – which is exactly where Rev A stood – and that is why it is written here rather than in a drawing note.

The shroud clears MP-01. The cable does not. At 12.50 mm the shroud leaves 5.50 mm of the 18.00 mm carrier-to-plate gap of ICD-EEG-006 section 4. The mated cable does not fit what is left: 8.50 mm of that gap is the socket, which leaves 9.50 mm for the housing and for the cable to turn, and **WH-01’s jacket is 4.30 mm OD with a 13 mm static bend radius**, so a 90-degree turn out of J14 needs 15.15 mm of height before the housing itself is counted. **MP-01 has no relief over either socket**: J14 runs x = 3.73 to 6.27 mm and the nearest slot edge in the plate is at x = 8.0 mm, so nothing over it is open at all, and over J30 there are four 12 x 3 mm jumper slots which give 3 mm of opening every 7 mm and no continuous exit. With the plate fitted the helmet cables cannot leave their sockets. MP-01 needs a relief over both; it is not cut in this revision and it is open item 18.

---

7. Routing inside the helmet

HM-01 is a monocoque with wiring channels subtracted at 1.9 mm radius, giving a 3.80 mm bore and 11.34 mm2 of channel area. Each section carries **two** parallel channels: **channel A** on the outer wall for WH-01, and **channel B** on the skull-facing inner wall for WH-02, separated by the section’s central web, minimum 6 mm centre to centre. That separation is the physical form of the ECO-EEG-014 rule that the light group stays away from the electrode group. Both channels are closed by snap-in cover strips on the skull-facing side, so conductors are laid in, not pulled; the fill limit is therefore set at **50 %**, and only a single replacement conductor is ever drawn through.

Two occipital entries, not one: **OE-1** for WH-01 (left of the midline) and **OE-2** for WH-02 (right of it). WH-BUS-01 sits at node **N1** immediately inside OE-2 under its own cover strip.

Segment	Channel	Conductors carried	Area (mm2)	Fill
OE-1 to N1, trunk	A	all 11 WH-01 signal conductors	4.24	37 %
OE-2 to N1, trunk	B	all 10 WH-02 conductors	4.42	39 %
N1 to crown NC, rear sagittal arch	A	Fz, Cz, Pz, C3, C4	1.93	17 %
N1 to crown NC, rear sagittal arch	B	LED1-5 + 5 LED_V tails	4.42	39 %
NC to Fz, forward sagittal arch	A / B	E_Fz / LED1 + tail	0.39 / 0.88	3 % / 8 %
NC to C3, coronal left	A / B	E_C3 / LED4 + tail	0.39 / 0.88	3 % / 8 %
NC to C4, coronal right	A / B	E_C4 / LED5 + tail	0.39 / 0.88	3 % / 8 %
N1 round the left halo	A	T7, F7, REF_L, BIAS_EL	1.54	14 %
N1 round the left halo	B	LED6, LED8 + 2 tails	1.77	16 %
N1 round the right halo	A	T8, REF_R	0.77	7 %
N1 round the right halo	B	LED7 + tail	0.88	8 %

Sites and where each conductor leaves its channel, per DSN-EEG-002 section 2:

10-20 site	Carried on	Assembly	WH-01 pin	WH-02 pin
Fz	sagittal arch, forward of the crown	HM-04 #1	J14.1	J30.1
Cz	crown junction of both arches	HM-04 #2	J14.2	J30.2
Pz	sagittal arch, aft of the crown	HM-04 #3	J14.3	J30.3
C3	coronal arch, left	HM-04 #4	J14.4	J30.4

10-20 site	Carried on	Assembly	WH-01 pin	WH-02 pin
C4	coronal arch, right	HM-04 #5	J14.5	J30.5
T7	halo stub above the left ear	HM-04 #6	J14.6	J30.6
T8	halo stub above the right ear	HM-04 #7	J14.7	J30.7
F7	halo stub, left front	HM-04 #8	J14.8	J30.8
Left ear	left temple exit, free lead	ear clip	J14.9	–
Right ear	right temple exit, free lead	ear clip	J14.10	–
Fpz	halo front, free lead to a coupler	disposable K4 pad on WH-10	J14.11	–

Three wire exits the released frame does not have. Conductors 9, 10 and 11 leave the channels at the two temples and at the halo front, and none of those is a channel mouth. `tools/mech_gen.py` defines exactly three mouths and all three are occipital: `HM01_N1_MOUTH` at (0.00, -95.38) in the shell roof and `HM01_HALO_MOUTH` at (+/-45.41, -82.41, -11.25). `CH_BORE` is a constant 3.80 mm, so there is no stepped mouth anywhere for a strain relief to react against either. HM-01 is carried over as an STL with no STEP and no parametric source (PARTS-EEG-019 OA-1), so these are three features that have to be added to a model nobody can regenerate. Sections 3.1.2.1 provision 4 and 3.1.3 state the consequence: no pull test is written against an anchorage that does not exist. Open items 26 and 29.

Where the boom detaches. In Phase 1 the boom does not route through the frame. The boom assembly clamps to the left temple mount and its own 1700 mm lead (WH-03B) runs outside the frame to the 3.5 mm jack on the POD-P1 panel; the participant-facing detach point is that plug. In Phase 2 the boom lead enters the left temple channel and the detach point becomes a temple-mounted jack. Either way the boom, the eight cups and the two ear clips are the parts that go into the ultrasonic bath at refurbishment; the frame is wiped and never immersed.

Cover strips. Every channel is closed by a snap-in strip on the skull-facing side, released with a plastic pick. The strips over N1 (WH-BUS-01), over the crown node NC and over each HM-04 base are separately removable so that a single conductor can be replaced without opening a whole arch.

---

8. Assembly sequence

Workmanship to **IPC/WHMA-A-620 class 2** for the cable assemblies, with **IPC-A-610 class 2** governing the carrier-end assembly they mate to. There are no splices anywhere in the kit: the drain is crimped directly into position 12, and WH-BUS-01 replaces what would otherwise have been eight LED_V splices.

#	Step	Tooling	Check before proceeding
1	Cut all conductors to the section 3 schedule, +10/-0 mm	calibrated tape, flush cutters	100 % length check on the first article, then 1 in 5
2	Strip 3.0 mm at the connector end, 4.0 mm at the site end	thermal or precision die stripper set for PTFE	no nicked, cut or missing strands; PTFE not drawn back
3	Crimp male contacts on the connector ends	Harwin Z20-320	visual to A-620 class 2: insulation crimp on insulation, conductor crimp on conductor, bell-mouth present, brush visible
4	Lay up WH-01: eleven conductors, binder tape, foil 25 % overlap, drain in contact with the foil, TPU jacket	jacketing die or heat-shrink jacket	OD 4.3 mm nominal, 4.6 mm maximum, measured on 3 points
5	Terminate the WH-01 drain: sleeve, crimp, insert in position 12	Z20-320, 2:1 clear heat-shrink	one drain, one contact, no stray strand outside the sleeve
6	Cut back and insulate the WH-01 screen at the helmet end, 10 mm, adhesive-lined shrink	hot-air, 120 C	screen not touching any conductor or any metal part
7	Lay up WH-02, ten conductors with LED_GND at the centre, TPU jacket	as step 4	OD 4.5 mm nominal
8	Insert contacts into the housings in the section 3 order	–	100 % pin-by-pin read-back against the wire list, by a second person
9	Build WH-03, WH-04, WH-05, WH-07 pigtails and the adapters WH-ADP-01/-01B/-02/-03 to section 3.9	as above	drains landed at the carrier end only; on WH-ADP-01 and -01B the lug-to-contact order proved against the jack in hand, not against the catalogue

#	Step	Tooling	Check before proceeding
10	Build WH-09: fit the panel receptacle and its gasket to WH-ADP-04, terminate the USB-B end, bond the braid at both ends of this cable only	as above	continuity of all four conductors and the braid; **no continuity from any WH-09 conductor or the braid to DGND, to any harness drain or to the WH-07 return**
11	**Electrical test of every loose assembly per section 9, before anything is fitted**	test set of section 9	full record; a failed harness is scrapped, not reworked at the crimp
12	Fit WH-BUS-01 at N1, square pad 9 facing OE-2; solder LED_V to pad 9, LED_GND to pad 10 and the eight tails to pads 1 to 8 in WH-02 conductor order (section 3.2.1)	temperature-controlled iron, 320 C, no-clean; strip 4.0 mm, tin, feed through from the legend side and solder both sides	LED_GND on its isolated pad and connected to nothing else; board the right way round
13	Lay WH-01 into channel A and WH-02 into channel B, site by site, from N1 outward	plastic pick	no conductor under tension with the frame opened to 62 cm; min bend radius respected at every node
14	Terminate at each HM-04 **per section 3.1.1**: solder the conductor to the HM-04A contact tail 8 mm outside the body, sleeve the joint in adhesive-lined shrink, dress it through the inboard slot opening, solder LED leads A and B with the marked lead to LEDn, and coil the 40 mm site service loop. **Not buildable at this issue: HM-04A, HM-05C, the LED seat and the bayonet run do not exist (open item 22)**	iron, adhesive-lined 2:1 shrink	continuity site to connector re-checked per site; the joint dry, clear of the gel port and clear of the spring

#	Step	Tooling	Check before proceeding
14a	Fit the eight electrode assemblies: HM-05C crown into the HM-05B spigot, cup into the carrier, K12 spring into the HM-04 seat, carrier into the bore with its lugs in the two entry slots, then the HM-09 key a quarter turn clockwise. Repeat for all eight	HM-09 service key	retention by hand – a straight pull of about 10 N does not release it (SVC-EEG-013 section 3 R10); the crown seated; the light window clear. **ASM-EEG-007 Rev B stage 3 has no such step and no stage-3 sign-off line for it. This document does not number an ASM step; the gap is open item 28**
14b	Fit the two ear-reference couplers to the temple tails per sections 3.1.2 and 3.1.2.1: sleeve the free tail over its whole 250 mm per section 4, crimp or solder the GY socket to conductor 9 and the PK socket to conductor 10, then **mate the K2 clips and leave them mated**	crimp or solder per the coupler maker	both clips part by hand without a tool and re-mate fully home; sleeve continuous from the screen cut-back to the coupler body; **the clips leave the bench mated** – PKG-EEG-015 and IFU-EEG-014 carry the packing rule and it is open item 34. **RULED, not signed, open item 23**
14c	Fit the bias coupler to conductor 11 per section 3.1.3: cut the tail to the F recorded at the fitting trial, sleeve it, crimp or solder the TQ socket, dress it at the halo front and leave WH-10 mated	as 14b	F recorded on the build record for this unit, not assumed from the drawing; the coupler mates and parts by hand; **no 15 N pull is applied to this joint** (section 3.1.3). **RULED, not signed, open items 26, 30 and 31**
15	Close all cover strips	–	no conductor trapped in a strip; strips flush
16	Fit the pod-end P-clips, coil the 120 mm service loops, bond the WH-KEY-01 shrouds over J14 and J30, mate J14 and J30	–	shroud seated on the socket and square; **the housing enters one way round only, checked by trying it the other way**; housings fully seated

#	Step	Tooling	Check before proceeding
17	Fit WH-ADP-04 to the POD-P1 aperture and plug WH-09 into the isolator module; fit its P-clip	–	gasket seated square; plug fully home; P-clip 40 mm behind the plug
18	Label every cable at both ends per section 10	thermal transfer printer	labels legible and correctly oriented
19	Repeat continuity and screen isolation as an as-built check, **and the H1 leg through each of the three mated couplers**	test set	matches the step 11 record; each coupler mated and re-read

9. Test

These limits are the single home for the harness electrical test. JIG-EEG-009 section 4.2 cites this table and does not restate it. Where JIG-EEG-009 Rev C gave different figures – a 500 V all-pairs sweep, a 1000 V AC 1 s dielectric withstand, a 10 N termination pull on 1 in 10 terminations – those are superseded by the table below. **There is no AC dielectric-withstand test on any harness assembly.** The only high-voltage tests in this document are the 500 V DC insulation-resistance measurements of H4 and H10, and both are insulation-resistance measurements, not hipots.

Every harness is tested as a loose assembly **before it is fitted to a carrier and before the helmet-end terminations to the contact-light LEDs are made**. This is not a preference. The 500 V DC insulation test would destroy the LEDs and would put 500 V on the ADS1299 module inputs through the 47 kOhm resistors. The order is: build the cable, test the cable, then terminate the ends that carry semiconductors.

ID	Test	Method	Limit	Applies to	Signed by
H1	Continuity, every conductor	4-wire milliohm meter, site terminal to its connector pin	1.0 Ohm maximum (WH-01, WH-02); 0.5 Ohm maximum (WH-07, WH-09)	all	harness operator
H2	Conductor-to-conductor isolation, all pairs	100 V DC insulation tester, every conductor against every other	100 MOhm minimum	WH-01 (66 pairs), WH-02 (45 pairs)	harness operator

ID	Test	Method	Limit	Applies to	Signed by
H3	Screen continuity and screen isolation	drain to J14.12; drain to every conductor; drain to each cup terminal (HM-04A and HM-05C, section 3.1.1) and to each HM-04 body. **The three touch-proof couplers are deliberately not in this test**: a DIN 42802 socket is single-pole and has no screen terminal, step 6 cuts the WH-01 screen back at the helmet end so all three tails are outside the screen, and the drain-to-every-conductor leg already covers them (section 3.1.2.1 provision 5)	drain to J14.12 2.0 Ohm maximum; all others 100 MOhm minimum	WH-01, WH-03, WH-04, WH-05	harness operator
H4	Insulation resistance	**500 V DC, 60 s**, conductor bundle to drain where a drain exists, and bundle to any exposed metal; **WH-02 has no screen or drain, so on WH-02 only the bundle-to-exposed-metal leg is run**. **New leg 2026-09-02: the three touch-proof couplers, plug withdrawn**, bundle to any exposed metal with the socket body and its shroud counted as exposed metal	100 MOhm minimum, no breakdown, no flashover	WH-01, WH-02, WH-03, WH-04, WH-05	QC inspector

ID	Test	Method	Limit	Applies to	Signed by
H5	Crimp pull-out	tensile tester, 10 s hold, on **3 sample crimps per batch**	**13 N minimum** for 28 AWG per IPC/WHMA-A-620 class 2	all crimped ends	QC inspector
H6	Assembly pull test	60 s hold	panel receptacle in its POD-P1 aperture 50 N minimum; boom detach 30 N minimum; HM-04 termination **15 N minimum** – which acts on HM-04A's anchorage in the body and on the solder joint to the conductor, **not** on the separable crown interface, which is designed to part at 3 to 6 N with the bayonet (section 3.1.1). **The three free-hanging couplers are excluded from the 15 N leg from 2026-09-02**: 15 N is defined against a body anchorage they do not have and sits above the 13 N at which H5 qualifies a 28 AWG crimp. They are not pulled at all until HM-01	WH-09, WH-03B, WH-01/-02 site ends	QC inspector

ID	Test	Method	Limit	Applies to	Signed by
			gains a drawn anchorage – sections 3.1.2.1 provision 4 and 3.1.3, open items 29 and 31		
H7	Flex	1000 cycles, +/-90 deg, at the crown transition	no continuity change; H1 repeated after	first article only	programme engineer
H8	Mate and de-mate	25 cycles on J14 and J30, then repeat H1. **New leg 2026-09-02: 100 cycles minimum on each of the three touch-proof couplers**, run as-built after step 15 with H1 repeated; the per-unit leg is step 19's as-built repeat and needs no new step	H1 limits still met; withdrawal 0.15 N minimum per contact; **coupler separation force inside 5 to 15 N at cycle 1 and at cycle 100**	first article only	programme engineer
H11	Finger-safety of the unmated couplers	**IEC 60601-1 / IEC 61032 test probe B** applied to each unmated touch-proof socket, all approach angles	no contact with any live part; recorded in the FAI pack	first article only, all three couplers	programme engineer, countersigned by the safety reviewer
H9	Wire-list verification	pin-by-pin read-back against section 3 by a second person	100 % match	all	second operator

ID	Test	Method	Limit	Applies to	Signed by
H10	WH-09 barrier isolation	500 V DC, 60 s, every WH-09 conductor and its braid against DGND, against every harness drain and against the WH-07 return, with WH-09 unplugged from the module	100 MOhm minimum	WH-09	QC inspector

The 500 V figure and why it is that number. No insulation-test value existed anywhere in package v1. 500 V DC is the standard insulation-resistance test voltage for equipment working below 50 V and is 200 times the +/-2.5 V that the electrode conductors actually see. A conductor that passes 100 MOhm at 500 V has, by the same insulation, a calculated leakage of 25 nA at the working voltage, 400 times below the 10 uA normal-condition limit of RFQ S-02. The test is a screen for damaged insulation and trapped strands, not a proof of the safety case; the safety case rests on R1 to R16 (E-07) and the ADuM4160 (E-24), and its single-fault arithmetic does not close today (section 5.5).

The same 500 V DC insulation-resistance measurement, made across the isolation barrier on the assembled unit, is the per-unit isolation test for the kit. The 2.5 kV RMS type test is the isolator supplier's certificate and is checked once at incoming inspection; there is no per-unit AC hipot station anywhere in this programme.

Record. Every measurement is recorded against the harness serial in the per-unit test record that ships with the kit, in the format of QP-EEG-010: harness part number and revision, serial, operator, date, instrument and its calibration due date, and every value in H1 to H6 and H10 as a number, not a pass mark. H7, H8 and H11 are first-article only and are recorded in the FAI pack. The record is signed by the harness operator, countersigned by the QC inspector, and the FAI pack is accepted by the programme engineer before the second unit is built.

TST-EEG-004 has no step that checks the harness record, and this document does not invent one. TST-EEG-004 Rev C owns the step numbers; what is needed is a document check on the pack – "harness record present and within limits" – not a repeat of the measurements. It is raised as open item 6 in section 11 against TST-EEG-004's next revision. Rev A of this document numbered it T2a, which it had no authority to do.

Estimated test time, calculated from the step count and not measured: 14 minutes per harness set, of which 6 are the all-pairs isolation sweep. This is additional to the per-unit test time of TST-EEG-004 section 10 and belongs in the RFQ pricing line for provisioning and functional test. The "25 minutes per unit" figure quoted in Rev A is withdrawn: it was arithmetically impossible against TST-EEG-004's own step durations, and TST-EEG-004 now carries the real figure.

Fixture: a 12-way and a 10-way mating adapter built from the same Harwin housings and contacts as the harness, so the jig proves the real mate and not a laboratory approximation. These adapters are fixture sub-assemblies and are numbered in JIG-EEG-009 under the FIX-01 to FIX-04 naming; this document does not number them.

10. Labelling

Every cable carries one label at each end, applied before the assembly leaves the bench.

Field	Format	Example
Part number and revision	`WH-EEG-008-nn RvB`	`WH-EEG-008-01 RvB`
Variant	`-P1` or `-H2`	`-P1`
Unit serial	`TIOV-B-nnnn`, as issued by PKG-EEG-015	`TIOV-B-0007`
End identifier	`A` = carrier end, `B` = helmet or panel end	`A`

The unit serial is programme prefix, hardware revision letter and four digits, and it is the same string on the label, in the Data Matrix, in the USB `iSerialNumber`, in the calibration record and on the packing list. Phase 1 uses 0001 to 0009.

Label material: printable heat-shrink polyolefin sleeve, Brady PermaSleeve PS-187-2-WT class or equivalent, thermal-transfer printed, 3.2 mm recovered diameter for the jacketed cables and 1.6 mm for the pigtails. Wrap-around adhesive labels are not accepted: the whole kit is wiped with 70 % IPA at every refurbishment and adhesive labels curl. The sleeve is positioned 25 mm behind the connector so it is readable with the connector mated.

Individual conductors are identified by colour, not by a printed number. The colour table of section 3 is the same table used for the HM-04 site labels, the POD-P1 panel legend required by RFQ A-06, and the runner's on-screen placement guide, so that the head, the panel and the screen cannot disagree about which site is which. That statement held for WH-01 and WH-02 only until this issue: **it now holds for the three EMG leads as well**, because section 3.6 is re-issued to the red / yellow / green ruling of IFU-EEG-014 Rev B section 13.2, so the lead in the participant's hand, the panel legend and the placement guide name one code.

Packing: each helmet ships with its harness already fitted inside HM-01, not loose. Spare harnesses ship coiled at 120 mm diameter, tied with two hook-and-loop straps, in an antistatic bag with the label visible through the bag.

Spares policy, to be costed in AVL-EEG-017 and the internal BOM, where the harness currently carries no cost at all: one spare WH-10 bias lead and two spare ear-reference couplers per ten kits, two spare WH-01/WH-02 sets and two spare boom assemblies per twenty-five kits, alongside the existing two spare frames, plus one spare EMG lead set and one spare WH-09 pigtail per ten kits. The participant's host and charge leads are the A-07 cables and are replaced from stock, not from this line.

11. Open items

An item that closes keeps its number and stays in this table with the text it carried, so that a reader who saw the earlier issue can tell a closed item from a deleted one. **Items**

7 and 15 are the closed rows at this issue; items 22 to 28 were new at it and items 29 to 34 are new on 2026-09-02 with the rulings of sections 3.1.2.1, 3.1.3 and 3.1.4.

#	Item	Who closes it
1	**No safety engineer has reviewed this design.** The single-fault case of section 5.5 – one electrode conductor shorted to the drain giving a calculated 53.2 uA against the 50 uA limit of S-02 – is a specific question for that review, together with ECO-EEG-024's 68 kOhm answer and the E-10 widening it forces. It does not block fabrication or quoting; it blocks use on a person.	RISK-EEG-011 safety review
2	**The isolator module presents USB-B where RFQ E-24 asks for USB-C.** WH-09 is the interim answer and is built for Phase 1; the non-conformance is open until a module with a USB-C host receptacle is qualified.	AVL-EEG-017, then an ECO
3	**J15 to J17 have no confirmed carrier part** (section 3.6). A touch-proof 1.5 mm socket with a PCB-mount signal pin and two 1.5 mm retention posts must be sourced and first-articled before Phase 2; AVL-EEG-017 carries a 12-week lead-time risk against it.	AVL-EEG-017, before Phase 2
4	**The boom preamplifier part is not chosen** (section 3.3). MAX9814 is AGC and is not approved; a MAX4466-class fixed-gain module is preferred; the interface in ICD-EEG-006 is what is specified until one is bought and measured.	AVL-EEG-017, before Phase 2
5	**No room-microphone module is known to meet E-15's hardware mute** (section 3.5). The fallback is a programme-designed capsule-plus-switch sub-assembly that has no drawing and no part number.	AVL-EEG-017, then a new drawing
6	TST-EEG-004 has no step that checks the harness test record is present and within limits. This document will not number one.	TST-EEG-004 next revision

#	Item	Who closes it
7	**Closed on 2026-09-02. The EMG lead colours are ruled red, yellow and green** by IFU-EEG-014 Rev B section 13.2, and section 3.6, the WH-06 table and section 10 of this document are re-issued to that code at this issue. *Was: “The EMG lead colours are not agreed. This document says white, brown and grey; IFU-EEG-014 section 13.2 and PKG-EEG-015 say red, yellow and green for the same three leads. The participant matches lead colour to site, so the two cannot both stand.”* The number is not reused. **What is still open is the change record, not the colour:** this correction has no entry in the ECO-EEG-016 register	Closed against IFU-EEG-014 Rev B section 13.2. The register entry is still owed by ECO-EEG-016
8	The charge-receptacle shell is specified isolated from DGND, which trades ESD and EMC robustness for leakage margin. Bonding it through an RC network was considered and rejected without measurement.	RISK-EEG-011, then an ECO if the ruling changes
9	LED_GND’s function as a guard conductor is a ruling, not a measurement. If the contact lights are changed to a common-return part, LED_GND becomes the return and this document is re-issued.	ECO-EEG-016
10	The contact-light bicolour phase driver is not implemented in firmware, so nothing yet exercises this cable as designed.	FW-EEG-001
11	Harwin ordering suffixes are given to the family and way-count. Confirm the current suffixes at IQC, and **with them the two housing dimensions WH-KEY-01 is cut for and the one it imposes**: body width across the flats, taken as 4.20 mm; male contact protrusion, taken as 4.00 mm; and a polarising rib at least 0.70 mm proud within the first way, without which the shroud keys nothing (section 6.1).	AVL-EEG-017

#	Item	Who closes it
12	WH-KEY-01, WH-BUS-01 and the adapters WH-ADP-01, -01B, -02, -03 and -04 have no line in the RFQ pricing template, so nobody is quoting them. All seven now have something to quote from – WH-BUS-01 fabrication data (section 3.2.1) and AVL-EEG-017 K42, WH-KEY-01 and WH-ADP-02/-03/-04 STEP and STL (sections 3.9 and 6.1) and AVL-EEG-017 K24, WH-ADP-01 and -01B a bought-part class and AVL-EEG-017 K37 – and the pricing template still has no line for any of them. **The same is now true of the harness itself**: AVL-EEG-017 K25 to K45 and section 1.6.1 price every material, connector, contact and tool in this document, and the pricing template has no line for those either.	RFQ-EEG-001 next revision
13	The Phase 2 in-frame boom route and the -H2 umbilical lengths are not dimensioned, because the occipital shell is a Phase 2 routing task and the shell in RFQ M-01 is sized for a carrier that no longer has those dimensions.	Phase 2
14	**Nothing in this package has been manufactured or measured, and no safety engineer has reviewed it.** Every length, OD, fill percentage, resistance, force and leakage figure above is calculated.	Phase 1 first article
15	**Closed on 2026-09-02.** PARTS-EEG-019 now records WH-BUS-01 as a two-layer board with the fabrication data set in `kicad/wh-bus-01/`, and AVL-EEG-017 K42 buys the bare board. *Was: "PARTS-EEG-019 Rev B's WH-BUS-01 entry is out of date in two places. It says single-layer, and the board is two layers because the pads are plated through holes; and it says 'specified, not built. No Gerber set has been generated for it', which section 3.2.1 and `kicad/wh-bus-01/` now answer."* The number is not reused. What is still true and is not an open item, because it is the state of the whole package: **no WH-BUS-01 has been fabricated.**	Closed against PARTS-EEG-019, 2026-09-02 issue

#	Item	Who closes it
16	**HM-01 has no geometry for WH-BUS-01 at node N1** – no pocket, no boss, no location dimension. The board is retained by the cover strip and its own solder joints, which is probably enough for a 14 x 10 x 0.8 mm part, but nothing says where within N1 it sits and no drawing can be checked against.	the mechanical package, before the first article
17	The IPC-D-356A netlist `tools/gerber.py` writes for EEG-CAR-01 states hole diameters in micrometres while its coordinates are in 0.0001 inch, so a tester reading the file to the standard sees every hole ten times too big. `tools/wh_bus.py` writes both fields in 0.0001 inch and says so in the file; the carrier's netlist is not corrected here.	DSN-EEG-003, then `tools/gerber.py`
18	**MP-01 has no opening over J14 or J30, so with the plate fitted the helmet cables cannot leave their sockets.** J14 is under the 8 mm solid border and J30 under four 12 x 3 mm jumper slots; 9.50 mm of the 18.00 mm carrier-to-plate gap is left above the socket, and WH-01's 13 mm static bend radius needs 15.15 mm for a 90-degree turn before the housing is counted (section 6.1). A relief over both sockets is not cut in this revision.	`tools/mech_gen.py` `mp01()`, then ICD-EEG-006 section 4
19	The J30 shroud reaches to y = 87.33 mm, **0.44 mm from J9's socket body**, which will not hold the end wall of the 1x4 module form of WH-KEY-01 that ICD-EEG-006 section 6.1 puts on J9. Either that form is open-ended on one side or J9 moves. The seventeen module-socket forms are not modelled.	ICD-EEG-006 section 6.1, then `tools/mech_gen.py`

#	Item	Who closes it
20	**Four of the five panel adapters sit in an opening that overlaps a button opening** (section 3.9): BTN_B is concentric with the boom opening, BTN_STOP contains the room-microphone port, BTN_A overlaps the headphone opening, and the charge opening runs into both. The adapters are dimensioned to the openings as specified; none can be fitted until `pod_base()` is corrected.	`tools/mech_gen.py pod_base()`, then RFQ M-02
21	**`pod_base()` carries no boss for WH-ADP-02, WH-ADP-03, WH-ADP-04 or the WH-09 P-clip**, so all three adapters are bonded to the wall and the section 3.8 P-clip has nothing to screw into. Test H6 pulls the panel receptacle at 50 N, which is then a pull on a bonded joint that has not been made or tested. Both USB plates carry M2.5 clearance holes for the screws that would replace the bond.	`tools/mech_gen.py pod_base()`
22	**The HM-04 termination is a proposal and not a design.** Section 3.1.1 specifies the joint, and HM-04A, HM-05C, the HM-04 anchorage, the LED seat, the dressed conductor exit and the circumferential bayonet run all have to be drawn and approved before the site end of either helmet cable can be built. `mech_gen.py hm05b()`'s flank solder-tag pocket is superseded by it and is withdrawn. **Narrowed on 2026-09-02: two of those six are cut.** The circumferential bayonet run exists and measures 0.000 mm3 of interference through the quarter turn and through the 0.40 mm of travel, and the LED seat exists as the outboard of two separated pockets, which also closes RISK-EEG-011 SF-9's shared cavity. **Still owed: HM-04A, HM-05C, the 15 N anchorage against a seat that now runs to z 15.60, the dressed inboard exit, and the LED's two lead passages out of its new blind pocket.**	The mechanical reviewer, then the safety reviewer, then AVL-EEG-017 K25 and K26

#	Item	Who closes it
23	**The two ear-reference terminations are a proposal.** Section 3.1.2 resolves the three-way disagreement between this document, AVL-EEG-017 K2 and SVC-EEG-013 R4 in favour of a free-hanging touch-proof coupler on the temple tail. The coupler has no vendor and the decision has no signature. **Ruled on 2026-09-02 (section 3.1.2.1) and still unsigned**: the coupler and the clip are unchanged; what is added is the packing rule (item 34), a stated 150 to 200 mm K2 lead with T8 re-measured, a sleeved free tail, a withdrawn strain relief (item 29), the H4 / H8 / H11 test legs and a 5 to 15 N separation window on K27.	The safety reviewer, then the programme lead for the packing rule, then AVL-EEG-017 K27 on the same purchase order as K2
24	**If item 23 is taken, the kit carries five 1.5 mm touch-proof interfaces and an EMG lead will mate an ear socket.** Not a safety fault – every one of them sits behind its own 47 kOhm resistor on the same rails – but a silent measurement fault that T10's lead-off check does not catch. Stated, not mitigated. **Grown on 2026-09-02 to six sockets and six plugs**, and it now runs in both directions: a K2 clip's plug will enter J15 to J17, and a K2 clip or a K3 EMG lead will enter the **bias** socket, putting an electrode on the driven output at a site RISK-EEG-011 works only at Fpz. Bounded by the same 47 kOhm as everything else since `design.py` lost its channel-11 override, and still a measurement fault. If the safety reviewer wants genuine non-interchangeability of the driven output, the answer is a connector class or a keyed shroud, not gender or colour.	RISK-EEG-011, then IFU-EEG-014's placement guide; the safety reviewer on the driven-output residual
25	**WH-03B has a flex duty and no cycle count**, because nothing in the package says how many times a boom is set in a unit's life. Test H7 is run at the gooseneck root for the WH-03B first article and the number comes from there.	Phase 1 first article

#	Item	Who closes it
26	**Conductor 11 lands on an “Fpz bias pad, solder tag” that has no part number, no drawing and no feature on any model.** It is the same defect as item 22 at a different site and section 3.1.1 does not close it. **Narrowed on 2026-09-02 by section 3.1.3, not closed**: the pad is deleted as a helmet feature and becomes a disposable K4 pad on the new WH-10 lead, and conductor 11 terminates in a TQ touch-proof socket of the K27 class. What is still owed on the frame is a **halo-front channel mouth and a dressed exit**, on a carried-over STL with no parametric source (PARTS-EEG-019 OA-1) – one of the three wire exits of section 7.	The mechanical reviewer, with item 22; then the safety reviewer, PARTS-EEG-019 and AVL-EEG-017 K47
27	**WH-03, WH-04, WH-05, WH-07 and WH-09 still have no row in `EEG_kit_BOM_for_bidders_RevC.xlsx`**, which is the sheet a contract manufacturer quotes from; WH-01 and WH-02 have rows that read “custom assembly / none”. AVL-EEG-017 section 4 now carries the material, connector, contact and tooling lines behind all eight, so the rows can be written; this document cannot write them.	The kit BOM, at its next issue
28	**ASM-EEG-007 Rev B stage 3 has no step that terminates at HM-04 and no step that fits the cups, the bayonet carriers or the springs**, and its section 9 stage-3 sign-off table has no line for either. Steps 14, 14a, 14b and 14c of section 8 above are the harness half; the ASM steps and their sign-off lines are not this document’s to number. ASM section 4.2’s “the window is on one side of the body only” is also wrong against the released HM-04 model – there are two openings and, since 2026-09-02, they are two separate pockets.	ASM-EEG-007, next revision

#	Item	Who closes it
29	**The ear couplers have no strain relief and no anchorage, and the H6 15 N leg is withdrawn from them.** The proposal's 2.0 mm adhesive-lined bulb at a temple channel mouth is deleted because the released HM-01 has no temple mouth – three mouths, all occipital – and `CH_BORE` is a constant 3.80 mm, so nothing reacts a bulb anywhere. Carried open beside the identical OE-1 / OE-2 anchor entry in KNOWN_ISSUES.txt. Nothing is pull-tested here until a temple wire exit is added to `tools/mech_gen.py` and registered.	The mechanical reviewer with the PARTS / MECH owner, then this document's section 9
30	**The free tail F on conductor 11 is not a number yet**, and its cut length is 1940 + F. It is a fitting dimension: bounded below by the clearance a hand needs to mate the coupler clear of the HM-02A brow pad, above by the 250 mm the ear tails carry. **Conductor 11's 1980 mm must not be carried forward** – it contains no free tail. Conductors 4 and 5 cut at the same figure for unrelated reasons and do not move.	The first fitting trial; recorded per unit on the build record at step 14c
31	**The bias coupler's retention value depends on a frame feature that does not exist.** With a drawn HM-01 anchorage it aligns with H6 at 15 N; without one it is capped at H5's 13 N minimum for a 28 AWG crimp, and 15 N must not be written against it.	The mechanical reviewer, with items 26 and 29
32	**The K12 spring has no retention or capture feature, no ASM fitting step and no SVC handling step.** Section 3.1.4 issues the spring itself; nothing holds it if a carrier is drawn without one, and a lost or mis-seated ring is a silent open electrode under a cup that still looks fitted. KNOWN_ISSUES.txt already records that no assembly step fits the cups, the carriers or the springs at all.	The mechanical reviewer for the capture feature; ASM-EEG-007 and SVC-EEG-013 for the steps, with item 28

#	Item	Who closes it
33	**The HM-04 / HM-05B rest datum is ambiguous by 0.10 mm, and it is the datum a spring is quoted against.** `mech_gen.py` states 0.40 mm of axial float from a 9.00 mm bore against an 8.60 mm body, which puts the spring's rest height at 3.50 mm; but the lug at z 1.20 comes to rest on a lip whose roof is at z 1.10, which stands the carrier 0.10 mm proud and makes the rest height 3.60 mm and the stroke 0.50 mm. Section 3.1.4's spring is specified to be indifferent to which is true; the models should not stay ambiguous.	`tools/mech_gen.py`, then MECH-EEG-020 sheet 8
34	**The packing rule that section 3.1.2.1's safety argument rests on belongs to four documents this one does not own.** The ear clips must travel mated and captive: PKG-EEG-015 section 1.1 line 2.1 to read "2 fitted, HELMET HM-01" and its (408, 162) 94 x 100 mm bay to become EMG LEADS only; IFU-EEG-014 section 1's table and section 9 step 1 re-issued; SVC-EEG-013 R12 to say the clips are left mated and dressed inside the helmet bay; RISK-EEG-011 H-20 and H-30 re-checked against the result. Without it the participant mates two of five identical DIN plugs every session, K27's life margin falls from 5x to 1.4x, and open item 24 becomes a per-session participant error.	The programme lead, across PKG-EEG-015, IFU-EEG-014, SVC-EEG-013 and RISK-EEG-011 together

12. Gaps closed by this document

Gap id	Closed by
harness-wire-list-absent	sections 3.1 to 3.8, one row per conductor with site, pin, net and colour
conductor-count-contradiction	section 2, with the v1 statements, the ECO-EEG-016 error and the Rev B budget that adds up
led-lines-have-no-driver	section 3.2 – J19 is 1x16 in Rev B and Q0 to Q7 reach LED1 to LED8 through R70 to R77 (ECO-EEG-001); the firmware driver is still open, item 10

Gap id	Closed by
contact-light-colour-vs-conductors	section 3.2, two-lead bicolour on LEDn / LED_V at 240 Hz with WH-BUS-01 splitting the common eight ways
screen-and-drain-termination	section 5, including the pod-end-only rule at R91 and the do-not-connect instruction at the head
cable-datasheet-missing	section 4
cut-length-schedule	sections 3.1 and 3.2, with the 1500 mm -P1 umbilical fixed
j14-mating-connector-and-tooling	section 6, including the mis-mate table and WH-KEY-01
emg-lead-assembly-and-din-screen	section 3.6, with deviation DEV-WH-01 recorded, the socket part still open, and the lead colour ruled red / yellow / green by IFU-EEG-014 Rev B section 13.2 and carried here
boom-cable-and-trrs-pinout	section 3.3
max9814-location-conflict	section 3.3 – the preamplifier is on MP-01, connected at J21, and which part it is remains open
panel-flylead-schedule	sections 3.4 and 3.5
host-cable-and-gland	**superseded.** The host connector is a socket on the isolator module presented through a gasketed POD-P1 aperture; WH-08 and the gland are withdrawn; section 3.8 specifies WH-09 and reconciles A-07
isolator-connector-mismatch	section 3.8 and open item 2 – stated as a live non-conformance with WH-09 as the interim answer
harness-test-specification	section 9, which is the single home for the limits that JIG-EEG-009 section 4.2 cites
harness-assembly-drawing-and-identity	part numbers WH-EEG-008-01 to -07 and -09, section 8 build order, section 10 labelling and spares
wh-key-01-has-no-geometry	section 6.1 and `tools/mech_gen.py wh_key01()` – three forms, released as STEP and STL, with the two housing dimensions it is cut for and the one it imposes named in the same section, and the one mis-mate it does not stop stated in the table
wh-adp-adapters-to-be-created	section 3.9 – five adapters, each with what it mates at both ends, what it is made from, its wire list and what has to be confirmed; WH-ADP-02, -03 and -04 modelled in `tools/mech_gen.py`, WH-ADP-01 and -01B specified as a bought jack class with no printed part

Gap id	Closed by
hm04-termination-does-not-exist	**not closed; specified.** Section 3.1.1 states what is missing, what the released HM-04 and HM-05B actually carry, what the joint has to do, a proposed method with its reasoning, the geometry the models must gain, the four alternatives rejected and who has to decide. It is a proposal and it is marked as one, and open item 22 carries it
ear-reference-termination-unbuildable	**not closed; specified, then ruled.** Section 3.1.2 sets out the three incompatible joints the package described and proposes the free-hanging touch-proof coupler that satisfies all three constraints, with the mis-mate it creates stated as open item 24. Section 3.1.2.1 rules it on 2026-09-02 and attaches nine provisions with values – the packing rule, a stated K2 lead length with T8 re-measured, a sleeved tail, a withdrawn strain relief, the H4 / H8 / H11 legs, a 5 to 15 N separation window, the colour, the life margin and the two-way mis-mate. It is not signed
bias-termination-does-not-exist	**not closed; specified.** Section 3.1.3 deletes the “Fpz bias pad” as a helmet feature, terminates conductor 11 in a TQ touch-proof socket of the K27 class at the halo front, adds WH-10 and a fourth K4 disposable pad, replaces the 1980 mm cut with 1940 + F, caps the pull test at H5’s 13 N, states the cross-mate residual, and says in the same paragraph as the current numbers that **SR-12 is not closed by any of it**. It is not signed
k12-spring-cannot-be-bought	**closed as a specification, open as a purchase.** Section 3.1.4 issues the spring against the deepened seat: 0.50 mm wire, 6.40 / 5.40 mm diameters, 2.2 active coils, 1.20 N/mm, 6.90 mm free length, 2.10 mm solid, 3.96 to 4.56 N over the working stroke and inside 3 to 6 N over the whole tolerance band, with the arithmetic, the no-gold relation, the R6 qualification regime and ten measurements on the first five. Five samples only; no fleet order; not signed
wh-03b-has-no-cable-specification	section 4, as a fifth column of the materials table, with the both-ends screen termination, the ultrasonic-immersion requirement, the corrected plug type and the flex duty stated in place of an invented cycle count

Gap id	Closed by
harness-has-no-purchasing-lines	AVL-EEG-017 section 4 lines K25 to K45 – raw cable, screen, drain, jacket, sleeving, connectors, contacts, tooling, panel jacks, receptacles and the WH-BUS-01 bare board – and section 1.6.1 for the per-way-count connector table. The bidders' BOM rows are still owed and are open item 27
harness-connector-system-decided-three-ways	section 6: one system, Harwin M20, chosen because `mech_gen.py` cuts the WH-KEY-01 cavity to it; ICD-EEG-006 section 3.1's Molex KK 254 wording is corrected and AVL-EEG-017 section 1.6.1 carries the per-way-count table
wh-bus-01-has-no-fabrication-data	section 3.2.1 and `kicad/wh-bus-01/` – two-layer Gerber X2, one Excellon drill file, an IPC-D-356A netlist, a placement and BOM note and a README, all written by `tools/wh_bus.py`. Nothing has been fabricated